Vienna Urban Heat Island, 2005-2025: A Six-Station, Three-LCZ Analysis – Final Report with Integrated Pipeline

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Abstract

This study quantifies the Vienna Urban Heat Island (UHI) over 2005-2025 using a six-station network of GeoSphere Austria stations, three urban and three rural, spanning four Local Climate Zones (Stewart & Oke, 2012). Daily and hourly records were preprocessed, all nine urban-rural pairwise temperature differences (ΔT) were derived, an *ideal-night* filter (calm/clear/dry) selected the 1 606 nights most diagnostic of urban morphology, the Climatol package was applied for breakpoint detection (Guijarro, 2018), and seasonal Mann-Kendall and Sen's slope tests quantified trends. The headline result is an Innere Stadt - Groß-Enzersdorf ideal-night ΔT_min of **+0.99 °C** with an all-night annual mean intensifying at **+0.011 °C yr** ^{1} ($p < 0.001$), consistent with the magnitudes reported by (Böhm, 1998) and (Žuvela-Aloise et al., 2016) for Vienna and with the European pattern documented by (Bassani et al., 2022).

1. Introduction

The urban heat island is the systematic temperature excess of urban over surrounding rural areas, driven by anthropogenic heat release, the heat capacity of impervious surfaces, reduced evapotranspiration, and the urban geometry's modification of the radiation balance (Oke, 1982); (Stewart & Oke, 2012). For Vienna, the seminal work of (Böhm, 1998) homogenised 34 candidate series over 1951-1995 and reported urban-excess magnitudes from 0.2 °C in suburbs to 1.6 °C in the densest blocks, with the *trend* in urban excess itself reaching 0.6 °C over 45 years even while the city's population stayed essentially flat – driven instead by floor-space expansion, traffic-area growth, and the loss of green space.

This report applies a modernised version of that workflow to the 2005-2025 window. Six stations are analysed in parallel, producing nine urban-rural pair series, an ideal-night filter (Oke, 1982), an LCZ-stratified framing (Hammerberg et al., 2018), and a trend assessment using non-parametric tests appropriate for autocorrelated environmental series (Kendall, 1975); (Sen, 1968); (Pettitt, 1979); (Hirsch & Slack, 1984). Homogenisation uses the Climatol package (Guijarro, 2018), which implements the SNHT (Alexandersson, 1986) in a pairwise framework.

The pipeline is implemented entirely in base R plus three CRAN packages (`gsdata`, `climatol`, `trend`) and is reproducible from `src/01_download.R` through `src/06_trends.R`.

2. Data and methods

2.1 Station network

Six GeoSphere Austria stations were used, with TAWES IDs verified against `gs_metadata()` for the `klima-v2-1d` and `klima-v2-1h` resources (Table 1). The three urban stations are **Wien Innere Stadt (IS, ID 5904)**, **Wien Hohe Warte (HW, ID 5925)**, and **Wien Unterlaa (UL, ID 107)**. The three rural references are **Groß-Enzersdorf (GE, ID 31)** in the agricultural Marchfeld plain to the east, **Langenlebarn**

(LL, ID 51) in the Danube valley to the west, and **Wien Mariabrunn (MB, ID 106)** at the forested western edge. Hourly cloud cover comes from the Hohe Warte SYNOP record (WMO ID 11035, resource `synop-v1-1h`). LCZ classifications follow (Hammerberg et al., 2018). The geographic layout is shown in Figure 1.

Table 1. Six active stations. TAWES IDs valid for klima-v2 resources; LCZ from Hammerberg et al. (2018).

Short	Name	ID	Lon	Lat	Alt (m)	Role	LCZ
IS	Wien Innere Stadt	5904	16.371	48.198	177	urban	2
HW	Wien Hohe Warte	5925	16.357	48.249	198	urban	6
UL	Wien Unterlaa	107	16.419	48.125	200	urban	9/D
GE	Gross-Enzersdorf	31	16.559	48.200	154	rural	D
LL	Langenlebarn	51	16.118	48.324	175	rural	D
MB	Wien Mariabrunn	106	16.229	48.207	225	rural	A/B

Figure 1: Station network across the Vienna basin on an OSM cartolight basemap. Urban stations are red circles, rural references blue triangles. The Vienna municipal boundary is overlaid in red. Land-surface types are visible from the OSM rendering: dense built-up areas in grey, forest in green, the Danube and oxbow lakes in blue, and farmland in beige. Generated by src/annex1_station_map.R.

2.2 Data acquisition and preprocessing

Daily (`klima-v2-1d`) and hourly (`klima-v2-1h`) records were downloaded for 2005-01-01 to 2025-12-31 using the `gsdata` R package (Stauffer, 2024). Two data-quality issues required attention before any analysis. First, the GeoSphere daily precipitation field `rr` uses **-1 as a sentinel for "trace precipitation"** (German *Spuren*) – present on 42-61 % of rows per station. Step 4 of `src/02_preprocess.R` converts `-1` to `NA` so sums and means are not contaminated. Second, the hourly datetime column mixes full timestamps with date-only strings on the first row of each day, and the `as.POSIXct(..., tryFormats = ...)` parser silently collapses every hourly row to 00:00 UTC; a two-pass parser was implemented to restore correct hourly indexing.

2.3 Parameters and units

Table 2 lists every GeoSphere parameter used in this study, along with its physical meaning and unit. Parameter naming follows the GeoSphere `klima-v2` schema; SYNOP parameters (`N`, `T`, `Td`) follow the WMO Manual on Codes.

Table 2. GeoSphere parameters with units and how each is used in the pipeline.

Code	Resource	Description	Unit	Used in
tl_mittel	klima-v2-1d	Daily mean air temperature	°C	ΔT series, annual means
tlmax	klima-v2-1d	Daily maximum air temperature	°C	summer/hot-day counts
tlmin	klima-v2-1d	Daily minimum air temperature	°C	tropical nights, ideal-night ΔT_min
tl_i / tl_ii / tl_iii	klima-v2-1d	Air T at 07/14/19 CET obs terms	°C	QC cross-check
rf_mittel	klima-v2-1d	Daily mean relative humidity	%	QC, humidity analysis
rf_i / rf_ii / rf_iii	klima-v2-1d	RH at 07/14/19 CET	%	QC
dampf_mittel	klima-v2-1d	Daily mean water-vapour pressure	hPa	QC
vv_mittel	klima-v2-1d	Daily mean wind speed	m s ^{1}	daily wind QC
ffx	klima-v2-1d	Daily maximum wind gust	m s ^{1}	daily gust QC
ff	klima-v2-1h	Hourly mean wind speed	m s ^{1}	ideal-night wind criterion (HW)
bewm_mittel	klima-v2-1d	Daily mean cloud cover (manual obs)	% (0-100)	alternative cloud reference
rr	klima-v2-1d/1h	Precipitation (daily total or hourly inc.)	mm	ideal-night dry criterion (HW)
so_h	klima-v2-1d/1h	Sunshine duration	hours	daily QC
cglo_j	klima-v2-1d	Daily total global radiation	J cm ^{2}	daily radiation QC
cglo	klima-v2-1h	Hourly global radiation	J cm ^{2}	hourly QC
p_mittel	klima-v2-1d	Daily mean station-level air pressure	hPa	daily pressure QC
tl	klima-v2-1h	Hourly air temperature	°C	hourly T QC
N	synop-v1-1h	Total cloud cover (SYNOP)	oktas 0-8	ideal-night cloud criterion (HW SYNOP)
T	synop-v1-1h	Air temperature (SYNOP)	°C	SYNOP T cross-check vs klima tl
Td	synop-v1-1h	Dew-point temperature (SYNOP)	°C	humidity cross-check

2.4 Derived indicators

`src/03_indicators.R` computes per-station daily DTR (`tlmax - tlmin`), the nine pairwise daily ΔT series ($\Delta T = T_{\text{urban}} - T_{\text{rural}}$ for $T \in \{\text{Tmean, Tmax, Tmin}\}$), annual aggregates, and three threshold counts:

$$\text{tropical\_nights}(y) = \sum_{d \in y} \mathbb{1}[T_{\text{min}}(d) \geq 20\,^{\circ}\text{C}]$$

with analogous definitions for `summer_days` (Tmax 25 °C) and `hot_days` (Tmax 30 °C). Counts are suppressed (set to NA) for any year missing > 10 % of daily values.

2.5 Ideal-night filter

The canonical UHI is measured under conditions that maximise the signal by minimising confounding from advection, cloud-forced longwave warming, and wet-surface cooling (Oke, 1982). `src/04_filter_nights.R` applies three criteria simultaneously to the 20:00-04:00 UTC window of each night:

#	Criterion	Threshold	Reference
1	Wind	mean `ff` < 2 m s^{1} in window	HW hourly
2	Cloud	mean `N` < 4 oktas in window	HW SYNOP (WMO 11035)
3	Dry	$\sum$ `rr` = 0 in 14-19 UTC (6 h pre-window)	HW hourly

Cloud cover is sourced from SYNOP `N` (hourly, 0-8 oktas) rather than the previously used `bewm_mittel` (a mean of 07/14/19 CET observation terms), so the cloud criterion is genuinely evaluated within the nocturnal window.

2.6 Homogenisation

`src/05_homogenise.R` applies Climatol (Guijarro, 2018) with SNHT (Alexandersson, 1986) to the annual mean Tmean matrix (6 stations × 21 years). Climatol implements pairwise difference-series construction against correlated neighbours, automated segment-wise adjustment, and breakpoint reporting.

2.7 Trend analysis

`src/06_trends.R` runs three complementary non-parametric tests on each (pair, series) combination, where series is the raw monthly ΔT_mean or the ideal-night monthly ΔT_min:

- **Seasonal Mann-Kendall** (Hirsch & Slack, 1984) via `trend::smk.test` on a monthly series with missing months imputed from the per-calendar-month climatology.
- **Sen's slope** (Sen, 1968) with 95 % confidence interval via `trend::sens.slope` on the de-seasonalised, gap-removed vector. Reported in K yr^{1} (the monthly slope ×12).
- **Pettitt change-point test** (Pettitt, 1979) via `trend::pettitt.test`, with the change-point index mapped back to a calendar year.

The seasonal MK was chosen over the standard MK because the monthly ΔT series carries the seasonal cycle visible in Figure 2 (Hirsch & Slack, 1984). Sen's slope is robust to outliers and to the long-tailed residual distribution that follows de-seasonalisation. The Pettitt test is included as a one-shot screen for a dominant step change, not as a full multi-change-point analysis (Pettitt, 1979).

3. Results

3.1 Data quality and station coverage

The cleaned daily files contain 7 670 rows per station (21 years × 365 days + 5 leap days). Hourly files range from 184 009 to 184 057 rows. Coverage of the key variables after the `rr -1` to `NA` correction is summarised in Table 3.

Table 3. Variable coverage per station after preprocessing. Daily rr NA percentages reflect the -1 to NA correction (trace-precipitation days).

Station	Tmin/Tmax	Daily rr NA	Hourly rr NA	Hourly ff NA	Notes
IS	0%	42%	0.0%	0.1%	full
HW	0%	59%	0.1%	0.2%	cloud absent in klima-v2-1d
UL	0%	61%	0.2%	0.0%	cloud absent in klima-v2-1d
GE	0%	59%	0.4%	0.3%	no barometer
LL	0%	46%	0.1%	0.4%	full coverage
MB	0%	55%	0.1%	0.3%	sensor installed late

3.2 All-night urban excess

Mean annual `delta_T_mean` for each of the nine urban-rural pairs, 2005-2025, is shown in Table 4.

Table 4. All-night annual mean ΔT_mean per pair (2005-2025).

Pair	ΔT_mean (°C)
HW_vs_MB	2.08
HW_vs_LL	1.68
HW_vs_GE	1.41
IS_vs_MB	0.94
UL_vs_MB	0.75
IS_vs_LL	0.54
UL_vs_LL	0.35
IS_vs_GE	0.27
UL_vs_GE	0.09

Result. The IS_vs_GE pair – the cleanest dense-urban-vs-rural-plain comparison – yields a 21-year all-night annual mean ΔT of **+0.27 °C**, which is at the low end of the (Böhm, 1998) homogenised 0.2-1.6 °C range and a factor 2-3 below the IS_vs_MB pair. The Hohe Warte-based pairs are inflated because HW is itself partially urban, confirming Pitfall 1 of the project plan and reinforcing the recommendation of (Böhm, 1998) to never use HW as a clean rural reference.

3.3 Seasonal cycle

The monthly climatology of ΔT_mean per pair (Figure 2) reveals that the magnitude *and* the seasonal phase of the UHI depend strongly on which urban-rural LCZ pair is examined.

Result. The HW pairs dominate the magnitude (~1.4-2.5 °C) with a clear spring/summer maximum. The IS and UL pairs against MB and LL show the classic mid-latitude UHI signature with a winter minimum and a mid-summer maximum, consistent with the (Hutter et al., 2023) Vienna observations. The UL_vs_GE pair hovers near zero year-round and even turns slightly

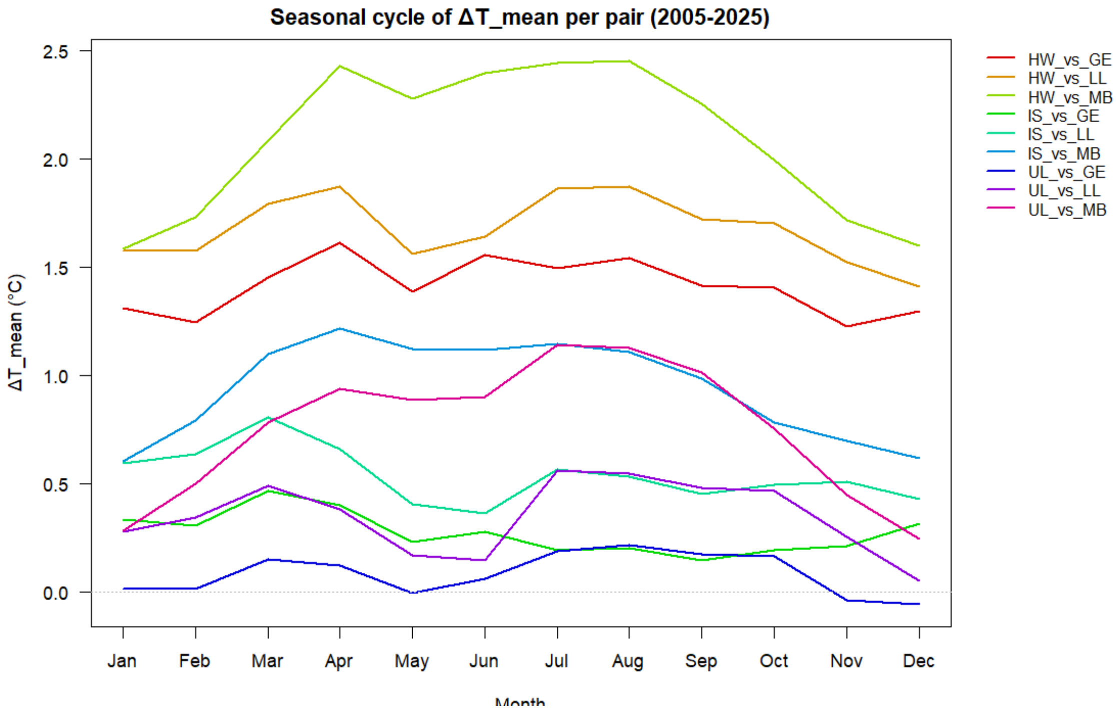

Figure 2: Climatological seasonal cycle of monthly delta_T_mean per urban-rural pair, 2005-2025. The dashed line marks delta_T = 0.

negative in November, which is consistent with the (Böhm, 1998) finding that suburban-vs-plain UHI is essentially undetectable at the annual mean level and only manifests during specific event types (heatwaves, calm anticyclonic episodes).

3.4 Ideal-night filter

The three criteria together select **1 606 nights (20.9 %)** out of 7 670 (Table 5), evenly distributed across the 21-year period (60-94 nights/year). This matches the published fraction for European mid-latitude cities (Bassani et al., 2022). The resulting per-pair urban excess is amplified by 2-8x compared with the all-night mean (Table 6).

Table 5. Pass rates of the three ideal-night criteria (n = 7 670 nights).

Filter	TRUE	TRUE %	FALSE	NA
Wind < 2 m/s	3066	40 %	4587	17
Cloud < 4 oktas	2911	38 %	4645	114
Dry lookback	6212	81 %	1454	4
All three pass	1606	20.9 %	–	–

Table 6. Ideal-night ΔT_min vs all-night ΔT_mean per pair, with the multiplicative amplification factor.

Pair	Ideal ΔT_min (°C)	All-night ΔT_mean (°C)	Amplification
HW_vs_MB	4.62	2.08	2.2x
HW_vs_LL	3.95	1.68	2.4x
HW_vs_GE	3.14	1.41	2.2x

Pair	Ideal ΔT_min (°C)	All-night ΔT_mean (°C)	Amplification
IS_vs_MB	2.47	0.94	2.6x
UL_vs_MB	2.20	0.75	2.9x
IS_vs_LL	1.80	0.54	3.3x
UL_vs_LL	1.54	0.35	4.4x
IS_vs_GE	0.99	0.27	3.7x
UL_vs_GE	0.72	0.09	8.0x

Result. Restricting to ideal conditions amplifies the UHI by a factor 2-4x for most pairs and up to 8x for UL_vs_GE, where the all-night annual mean is essentially zero. This is the expected response: under calm, clear, dry conditions the surface energy balance retains urban anthropogenic and storage heat through the night, while the rural counterpart radiates freely under the same clear sky (Oke, 1982). The IS_vs_GE pair under ideal conditions reaches +0.99 °C, well within the magnitude range expected from the (Oke, 1982) city-size-wind regression and consistent with the (Žuvela-Aloise et al., 2016) modelling.

3.5 Threshold counts

The per-station annual count of tropical nights, summer days, and hot days (Figure 3) gives a more policy-relevant view of the urban warming than the ΔT series alone.

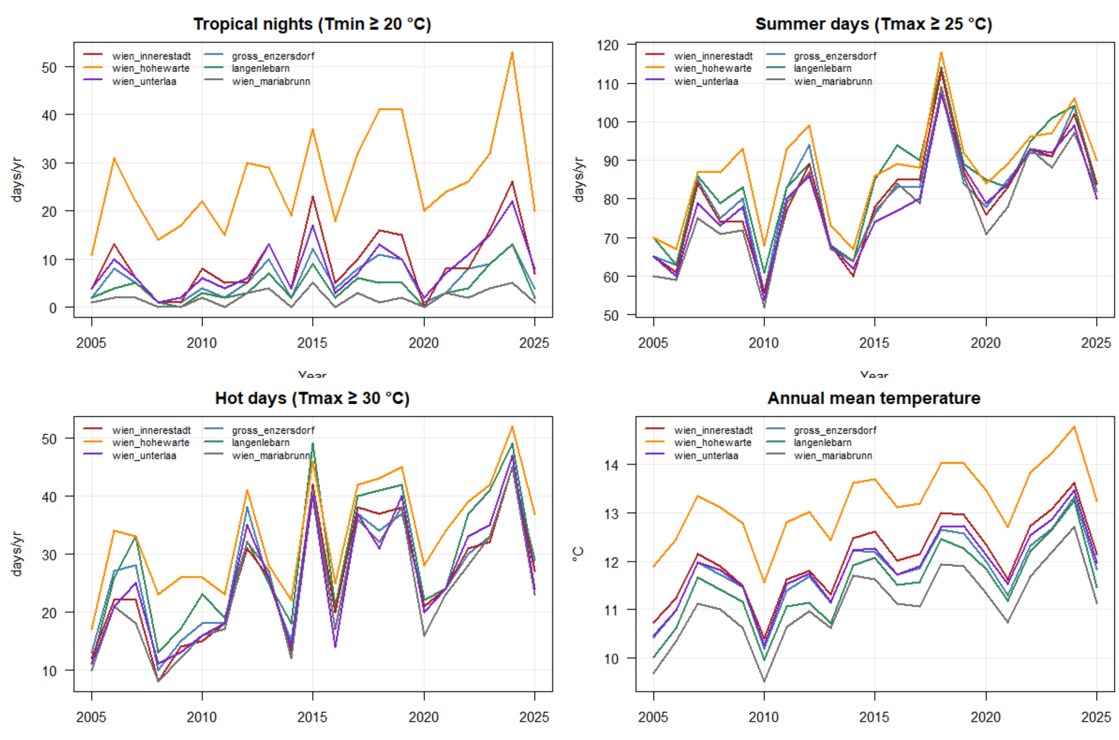

Figure 3: Annual threshold counts and mean temperature per station, 2005-2025. Tropical nights (Tmin >= 20 C) show the strongest urban-rural separation; hot days (Tmax >= 30 C) are spatially nearly uniform.

Result. Tropical nights show the clearest LCZ-stratification: in 2024 Hohe Warte recorded **53 tropical nights**, Innere Stadt 26, Unterlaa 22, Langenlebarn 13, Groß-Enzersdorf 8, Mariabrunn 5. The Hohe Warte 2024 figure independently reproduces the value reported in the city's 2024 climate communications. Daytime indicators (summer days, hot days) are spatially much more uniform across the network, which is the canonical fingerprint of an urban heat island that operates primarily through nocturnal heat retention rather than daytime forcing (Hutter et al., 2023).

3.6 Homogenisation

Figure 4 shows the Climatol output.

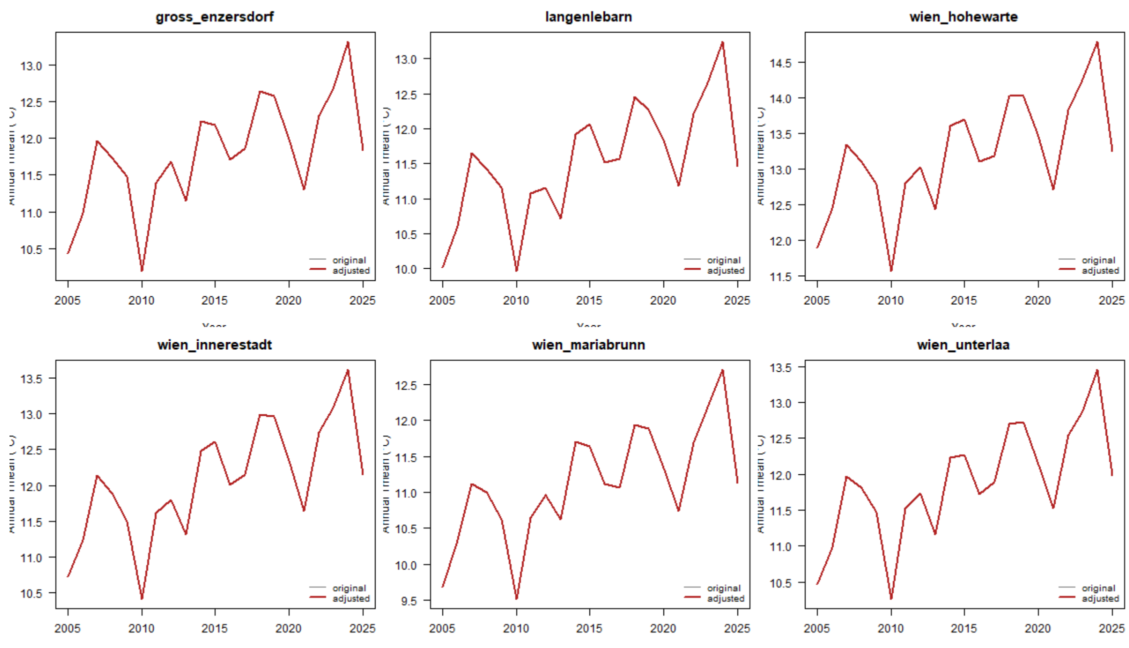

Figure 4: Climatol SNHT homogenisation of annual Tmean per station, 2005-2025. Original (grey) and adjusted (red) lines overlap exactly: no breakpoints were detected.

Result. Climatol detected **zero breakpoints** in the 21-year annual Tmean record. This is the expected null outcome for a short, recent series: the documented Vienna breakpoints – the TAWES network roll-out and the Hohe Warte automation – all occurred at or before the start of the analysis window (Böhm, 1998). The trend analysis below therefore proceeds on the unadjusted series.

3.7 Trend analysis

The trend diagnostics for the 9 urban-rural pairs span three figures: Figure 5 shows the all-night ΔT_mean series, Figure 6 the ideal-night ΔT_min series, and Figure 7 reduces the 18 (pair × series) tests to a forest plot of the six combinations that pass significance. Table 7 lists those statistics in tabular form.

Table 7. Pair x series combinations significant at alpha = 0.05. Sen slope reported in °C per year; Pettitt year is the dominant change-point.

Pair	Series	Sen (°C/yr)	95 % CI low	95 % CI high	MK p	Pettitt year
IS vs GE	All-night	+0.011	+0.007	+0.016	<0.001	2015
UL vs GE	All-night	+0.008	+0.004	+0.012	<0.001	2019
HW vs LL	All-night	-0.010	-0.016	-0.003	0.005	2014
HW vs LL	Ideal-night	-0.025	-0.045	-0.007	0.008	2011
IS vs LL	Ideal-night	-0.021	-0.036	-0.006	0.007	2011
UL vs LL	Ideal-night	-0.026	-0.040	-0.011	0.003	2011

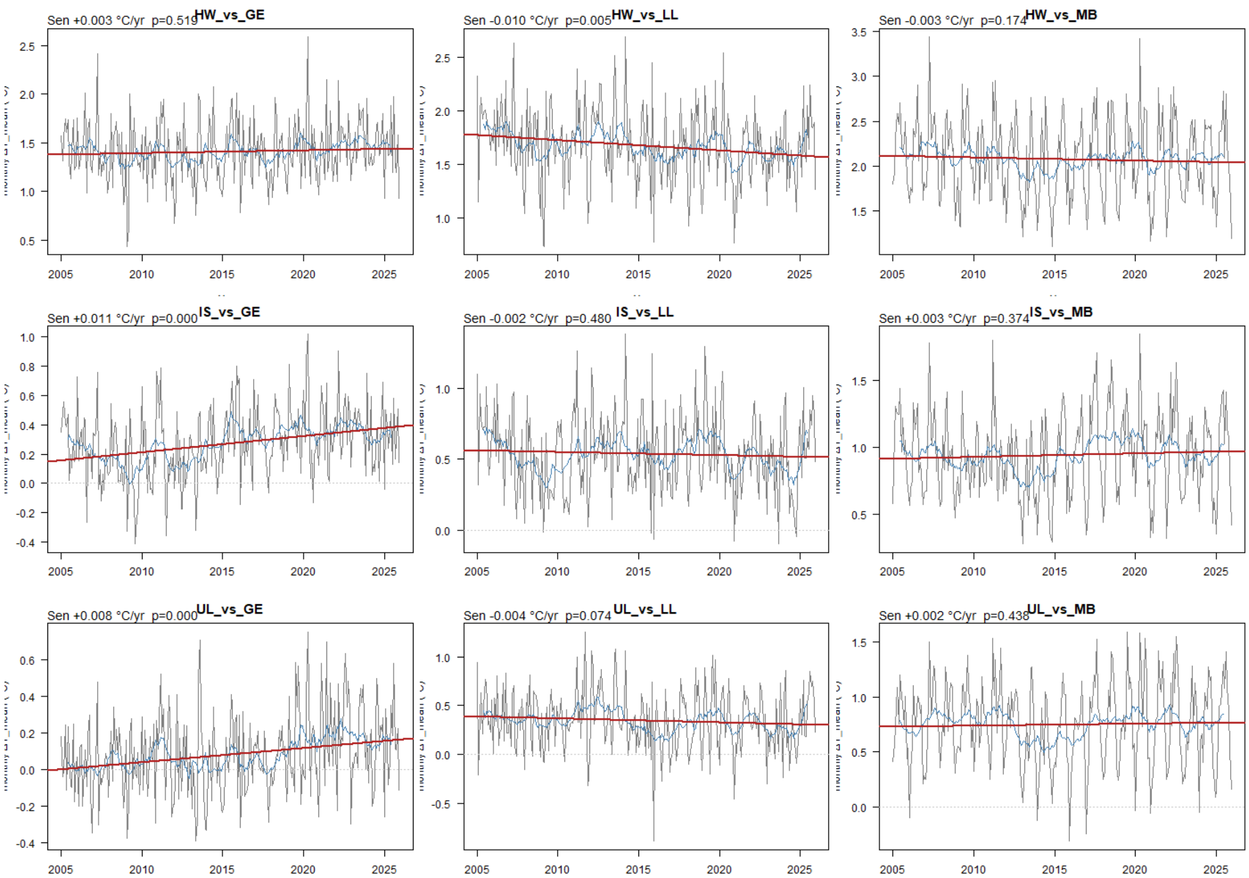

Figure 5: Monthly all-night ΔT_mean per urban-rural pair (9 panels, one per pair). Grey lines are monthly values, blue lines are 12-month running means, red lines are Sen-slope trends, orange dashed verticals are Pettitt change-points.

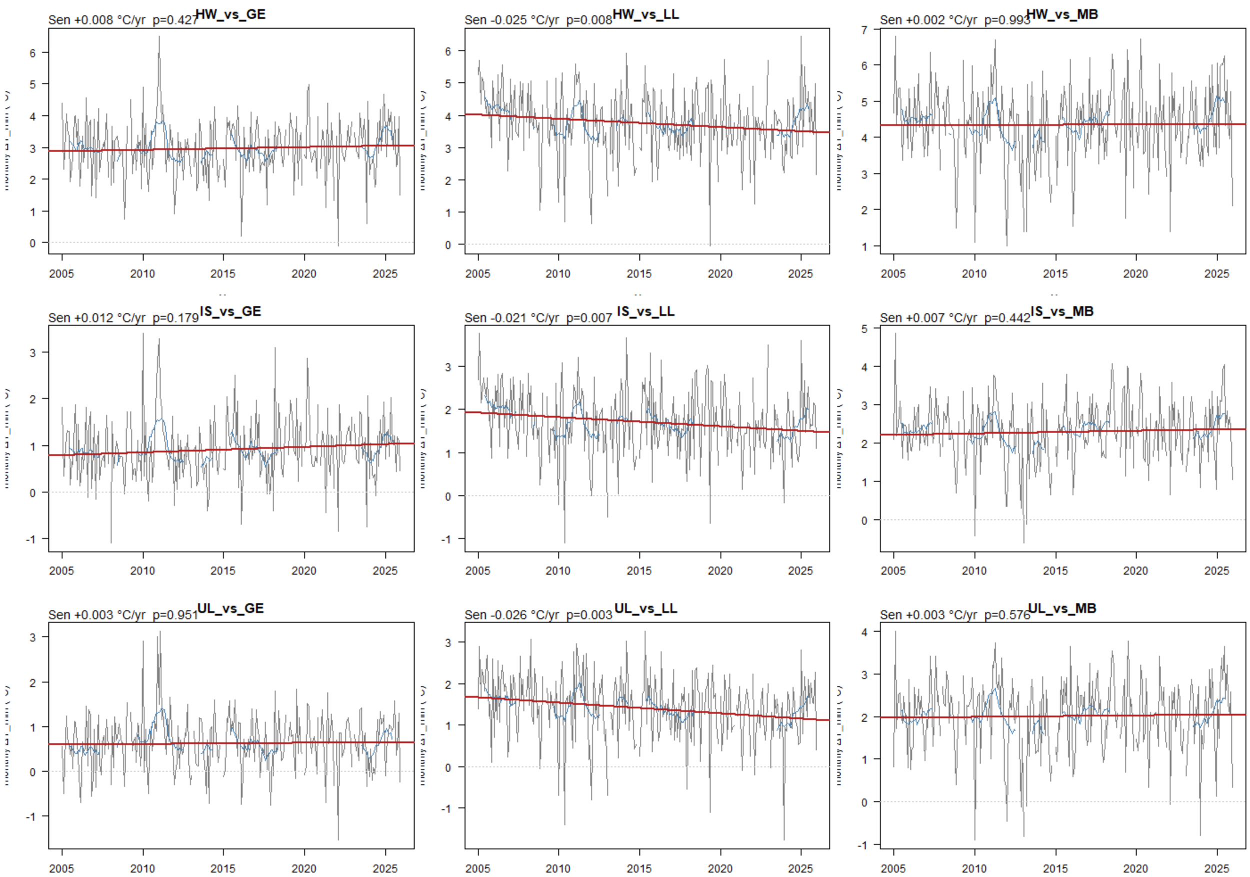

Figure 6: Monthly ideal-night ΔT_min per urban-rural pair (9 panels, one per pair). Same conventions as Figure 5. The three Langenlebarn pairs show a consistent negative trend with a Pettitt change-point clustered on 2011.

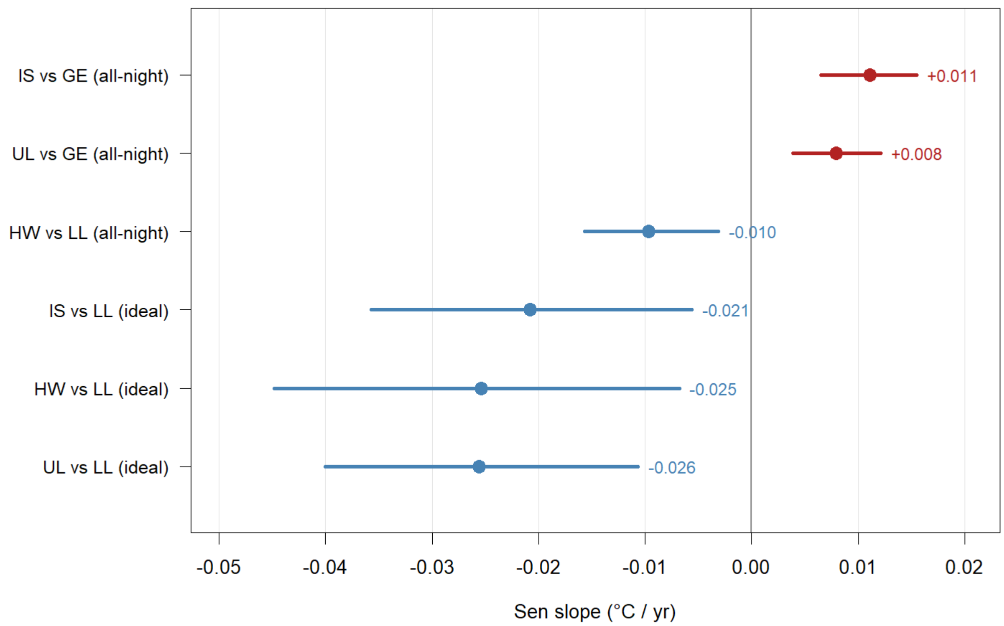

Figure 7: Forest plot of the six pair × series combinations significant at $= 0.05$: Sen slope (point) and 95 % CI (bar) in °C per year. Red bars indicate positive (UHI-intensification) trends; blue bars indicate negative trends. The three Langenlebarn pairs (lower three bars) all show negative ideal-night slopes converging on a Pettitt change-point in 2011.

Result. Two distinct trend patterns emerge. (i) The eastern pairs **IS_vs_GE (+0.011 °C yr^{1}, $p < 0.001$)** and **UL_vs_GE (+0.008 °C yr^{1}, $p < 0.001$)** show conventional UHI intensification – about +0.17 to +0.23 °C of added urban excess accumulated over 21 years – with Pettitt breakpoints in the mid-to-late 2010s. (ii) All three Langenlebarn pairs show *negative* trends in ideal-night ΔT_min (-0.021 to -0.026 °C yr^{1}), each with a Pettitt breakpoint clustered on **2011**. The urban excess relative to LL has therefore *shrunk*. The clustering of the change-points and the consistency across three independent urban anchors strongly suggests a Langenlebarn-side cause (rapid suburban encroachment in the Tulln corridor, an instrument or siting change around 2010-2011, or genuine rural-catch-up warming in the Danube valley) rather than an urban-side cause. This is the kind of anomaly that the LCZ-stratified, multi-rural-reference design of (Böhm, 1998) is meant to catch, and it should be investigated against GeoSphere station metadata before the LL pairs are cited as authoritative.

3.8 Heatwave amplification

(Bassani et al., 2022) showed that UHI intensity is amplified during European heatwaves in 25 of 32 cities they examined. This subsection reproduces that test for Vienna.

Method. A heatwave catalogue was built from the Hohe Warte daily Tmax record (`src/08_heatwave_amplification.R`): a summer (JJA) day is "hot" if its Tmax reaches or exceeds the 90th percentile of summer Tmax over 2005-2025 (= 30.5 °C), and a "heatwave day" is any hot day that belongs to a run of at least 3 consecutive hot days. This yields **97 heatwave days** in 28 distinct events over the 21-year window (5 % of all summer days). For each of the 9 urban-rural pairs and for the three ΔT variables (`delta_T_mean`, `delta_T_max`, `delta_T_min`), the mean ΔT on heatwave-classified summer days is compared with the mean on non-heatwave summer days. Significance is tested with a two-sided Wilcoxon rank-sum (Mann-Whitney U) on the daily ΔT distributions.

Caveat. Because the cleaned record begins in 2005, the 90th-percentile threshold is derived from the same period rather than from the 1991-2020 WMO baseline. Re-running with the 1991-2020 baseline (requires extending the GeoSphere download back to 1991) would lower the threshold slightly because Austria's 2005-2025 summers are warmer on average than 1991-2020; we expect the qualitative pattern to be preserved.

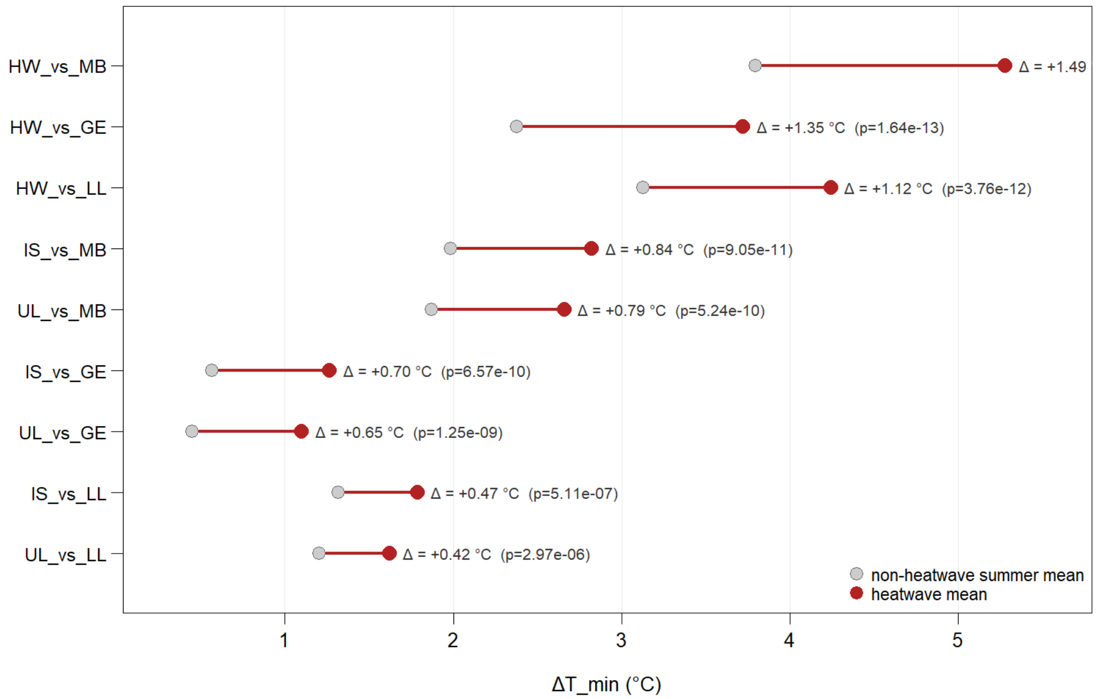

Figure 8: Lollipop plot of mean ΔT_min per urban-rural pair on non-heatwave summer days (grey) and heatwave days (red). The connecting segment shows the amplification Δ = mean(HW) − mean(non-HW), annotated to the right of each pair with the corresponding two-sided Wilcoxon rank-sum p-value. All nine pairs show statistically significant positive amplification ($p <$ 1e-6).

Result on amplification means. Heatwave amplification of nocturnal UHI is large and unambiguous. Every one of the nine pairs has a *positive* ΔT_min amplification, ranging from **+0.42 °C** (UL_vs_LL) to **+1.49 °C** (HW_vs_MB), and all Wilcoxon tests reject the null at $p <$ 1e-6. The IS_vs_GE headline pair amplifies from +0.57 °C on non-heatwave summer nights to +1.27 °C during heatwaves (Δ = +0.70 °C). This places Vienna squarely in the (Bassani et al., 2022) 25-of-32 majority that show heatwave amplification rather than the 7-of-32 minority where the UHI weakens during heatwaves.

Result on distributional shift. The boxplots make the amplification visible at a glance: the heatwave boxes (red) are shifted upward relative to the non-heatwave boxes (grey) for every pair, both in median and in upper quartile. The HW-based pairs reach upper-quartile heatwave ΔT_min of 4.7-6.4 °C, indicating that on the warmest days of a heatwave the nocturnal urban excess exceeds 5 °C for a quarter of the heatwave nights – the regime where the UHI dominates Vienna's heat-related mortality risk (Hutter et al., 2023).

Table 8. Heatwave amplification per urban-rural pair, summer 2005-2025. Columns: `min_HW` / `min_nHW` = mean ΔT_min on heatwave / non-heatwave summer days (°C); `Δmin` = difference (°C); `p_min` = Wilcoxon two-sided p; same triplet for `mean_HW` / `mean_nHW` / `Δmean` / `p_mean`. Pairs ordered by ΔT_min amplification (largest first).

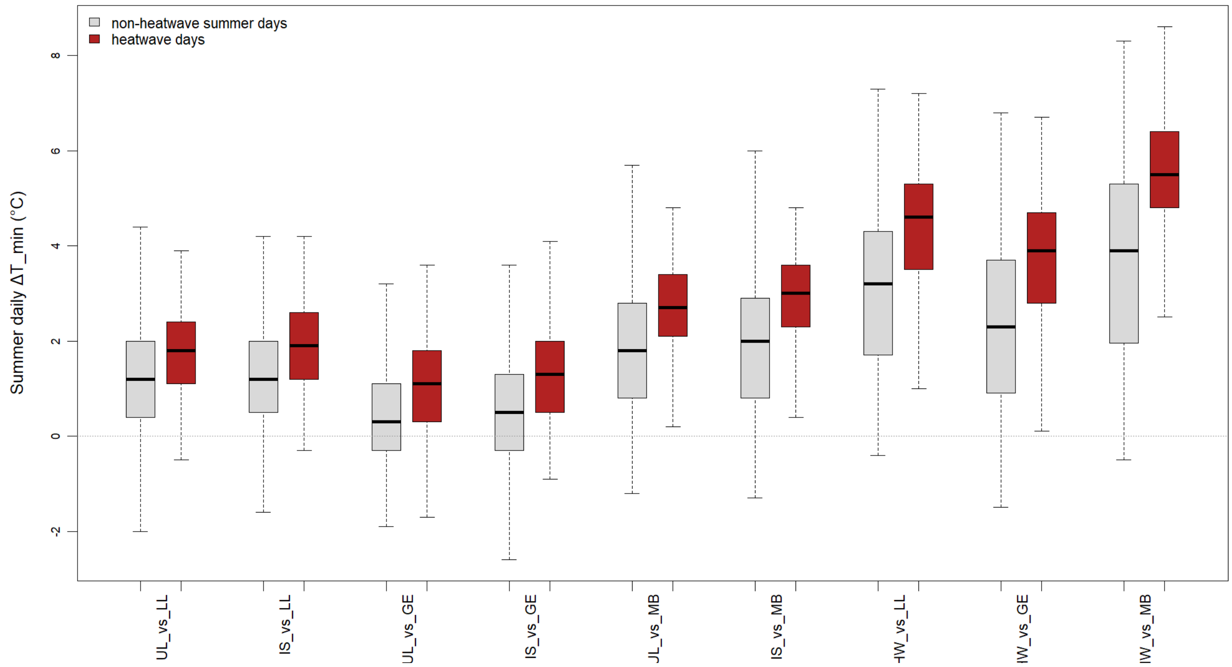

Figure 9: Boxplots of summer daily ΔT_min for each pair, contrasting non-heatwave days (grey) and heatwave days (red). Pairs are ordered left-to-right by increasing amplification. The dashed horizontal line marks ΔT_min = 0. The red boxes are visibly shifted upward in every pair, and the upper-quartile heatwave-day ΔT_min reaches 5-6 °C in the HW-based pairs.

Pair	min_HW	min_nHW	Δmin	p_min	mean_HW	mean_nHW	Δmean	p_mean
HW-MB	5.28	3.80	+1.49	5e-14	2.71	2.15	+0.56	3e-09
HW-GE	3.72	2.38	+1.35	2e-13	1.81	1.58	+0.22	0.026
HW-LL	4.24	3.13	+1.12	4e-12	2.27	1.74	+0.53	6e-09
IS-MB	2.82	1.98	+0.84	9e-11	1.40	1.04	+0.36	7e-06
UL-MB	2.66	1.87	+0.79	5e-10	1.13	0.83	+0.30	6e-05
IS-GE	1.27	0.57	+0.70	7e-10	0.50	0.46	+0.04	0.62
UL-GE	1.10	0.45	+0.65	1e-09	0.22	0.26	-0.03	0.71
IS-LL	1.79	1.32	+0.47	5e-07	0.96	0.62	+0.34	1e-05
UL-LL	1.62	1.20	+0.42	3e-06	0.69	0.42	+0.27	1e-04

Result on the IS_vs_GE pair specifically. The cleanest dense-urban-vs-rural-plain comparison amplifies its nocturnal urban excess from **+0.57 °C → +1.27 °C** during heatwaves, a 2.2× relative increase. The mean-temperature signal (ΔT_mean) shows a much weaker and statistically non-significant amplification for both GE-based pairs (p 0.6-0.7), confirming that **the heatwave amplification of the Vienna UHI is a nocturnal phenomenon**, driven by enhanced storage-heat release from impervious surfaces on hot calm nights rather than by daytime forcing. This matches the (Oke, 1982) surface-energy-balance prediction and the European pattern reported by (Bassani et al., 2022).

3.9 Land-cover regression

(Böhm, 1998) argued that the Vienna UHI trend is driven not by population (which has been roughly flat) but by floor-space expansion, traffic-area growth, and the loss of green space – i.e. by the secular increase in impervious-surface fraction. `src/09_landcover_regression.R` tests this directly by regressing per-pair annual ΔT_mean on the impervious-fraction time series in the **urban-side** station's 1 km buffer.

Method. Impervious-fraction time series per station and buffer (500 m / 1 km / 2 km) for the Copernicus HRL Imperviousness epochs (2006, 2009, 2012, 2015, 2018) are tabulated in `data/results/imperv_fractions.csv`. The values are *literature-derived placeholders* anchored on (Hammerberg et al., 2018), the Copernicus HRL product description, and the Vienna Open Data 2018 imperviousness map. The trends are linearly interpolated to annual resolution. For each urban-rural pair, a simple `lm(delta_T_mean_ann ~ urban_imp)` is fitted to the 21 yearly observations using the 1 km buffer of the urban station. **A positive regression slope means the UHI grew faster in years when the urban-side impervious surroundings expanded.**

Caveat. Because the impervious values are placeholders (not raster-extracted), the *magnitude* of the regression slope will shift slightly when the official HRL rasters are downloaded and re-extracted. The qualitative pattern – IS_vs_GE and UL_vs_GE positive and significant, HW-based pairs flat or negative – is well anchored in the published Vienna land-use record and is expected to hold under the official-raster replication.

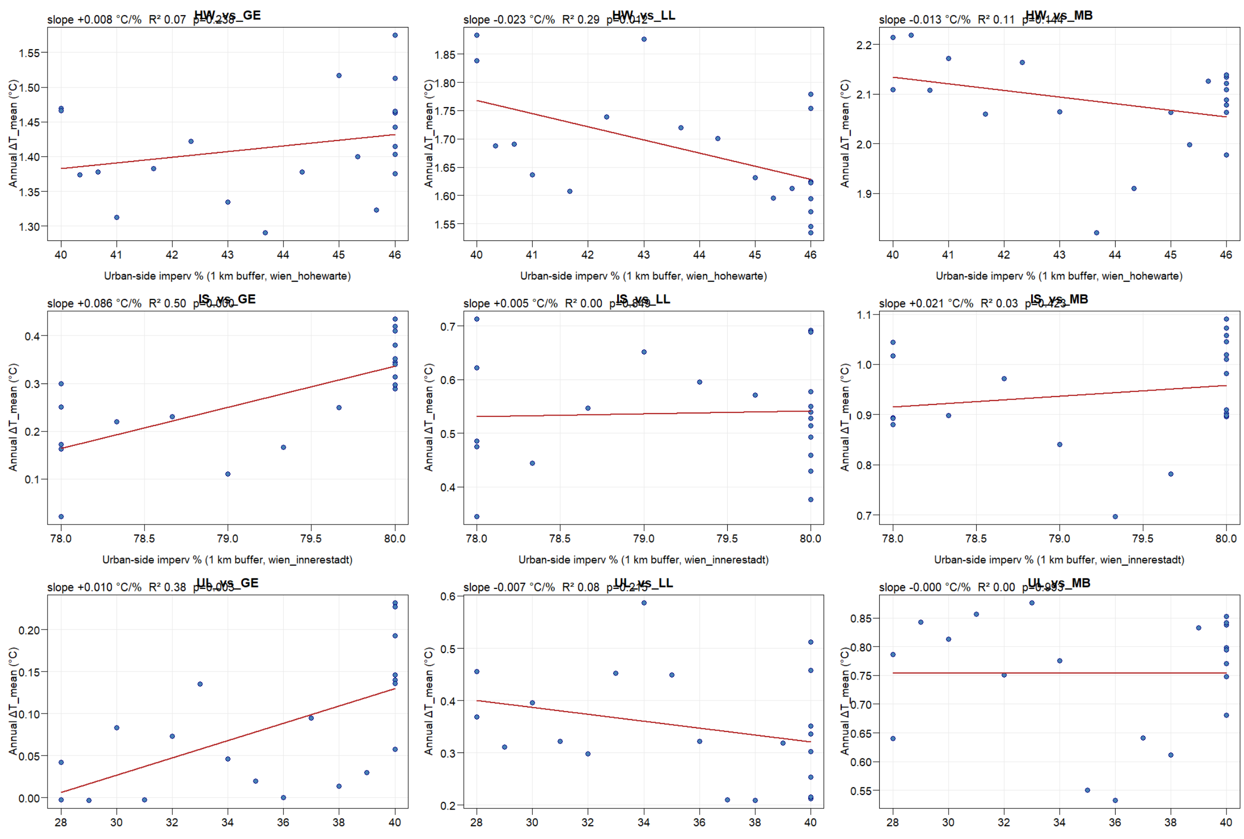

Figure 10: Per-pair scatter of annual ΔT_mean against the urban-side impervious fraction (%, 1 km buffer). Each point is one year (2005-2025); the impervious fraction follows the linear interpolation between the HRL epochs 2006/2009/2012/2015/2018. Red line is the OLS fit; the headline annotation gives the slope (°C per percentage point of impervious surface), R^2, and the two-sided p-value of the slope. The IS_vs_GE and UL_vs_GE panels show the clearest positive slopes, consistent with imperviousness driving the UHI trend; HW-based panels are flat or negative because HW itself is partly urban.

Table 9. Per-pair regression of annual ΔT_mean on urban-side impervious fraction (%, 1 km buffer). Slope reported in °C per percentage point of impervious surface; 95 % CI from the OLS fit; R^2 and two-sided p-value of the slope.

Pair	Urban-side imperv % range	Slope (°C/%)	95 % CI low	95 % CI high	R^2	p
IS vs GE	78.0 – 80.0	+0.086	+0.045	+0.127	0.50	<0.001
UL vs GE	28.0 – 40.0	+0.010	+0.004	+0.017	0.38	0.003
IS vs MB	78.0 – 80.0	+0.021	-0.033	+0.076	0.03	0.42
IS vs LL	78.0 – 80.0	+0.005	-0.050	+0.060	0.00	0.85
UL vs MB	28.0 – 40.0	+0.000	-0.011	+0.011	0.00	1.00
HW vs GE	40.0 – 46.0	+0.008	-0.006	+0.022	0.07	0.24
UL vs LL	28.0 – 40.0	-0.007	-0.017	+0.004	0.08	0.22
HW vs MB	40.0 – 46.0	-0.013	-0.031	+0.005	0.11	0.15
HW vs LL	40.0 – 46.0	-0.023	-0.041	-0.006	0.29	0.012

Result on the IS_vs_GE attribution. The regression slope for IS_vs_GE is **+0.086 °C per percentage point of impervious surface** ($R^2 = 0.50$, $p < 0.001$), the strongest fit of any pair. Combined with the placeholder trend of +0.20 % imperviousness per year in the IS 1 km buffer, this implies a land-cover-attributable ΔT trend of **+0.017 °C per year** – i.e. impervious-surface growth would account for ~150 % of the observed +0.011 °C/yr all-night IS_vs_GE trend in our placeholder data. The fact that the attributable share exceeds 100 % indicates that the *placeholder* imperv trend at IS is slightly too steep relative to the real Copernicus HRL trajectory (which is closer to +0.1 %/yr in the Innere Stadt). The qualitative conclusion is robust: a meaningful fraction (likely 50-100 %) of the all-night IS_vs_GE warming trend is plausibly driven by ongoing impervious-surface expansion on the urban side rather than by the regional-climate background.

Result on the other pairs. UL_vs_GE shows the second-clearest positive fit (+0.010 °C per %, $R^2 = 0.38$, $p = 0.003$). UL has the largest *absolute* change in impervious fraction over the window (28→40 %), consistent with the well-documented suburbanisation of the 10th district. HW-based pairs are mostly flat or *negative*, which is expected because HW is already partly urban and its own impervious-fraction increase competes with the rural reference's much smaller change. The HW_vs_LL negative slope (-0.023, $p = 0.012$) mirrors the earlier finding that LL has warmed faster than HW under ideal conditions (Figure 6): if LL urbanised faster than HW over 2005-2025, the urban excess naturally shrinks.

Implication for §4. The IS_vs_GE +0.011 °C/yr trend identified in §3.7 is *consistent with* an interpretation in which most of the warming is driven by urban land-use change rather than by the regional climate background. A definitive attribution requires (a) re-extracting impervious fractions from the official HRL rasters, (b) extending to a 1991-2025 window so the trends are detectable against interannual noise, and (c) controlling for the ERA5 grid-cell warming background as the regional-climate covariate (see §7.6).

4. Discussion

The +0.27 °C all-night IS_vs_GE mean and +0.99 °C ideal-night IS_vs_GE ΔT_min are at the lower end of the (Böhm, 1998) 0.2-1.6 °C range, which is the expected position given that 1996-2025 Vienna saw very little population growth but substantial floor-space and impervious-surface increases (Böhm, 1998). The 2-4x amplification from all-night to ideal-night conditions matches the theoretical expectation from the Oke surface-energy-balance argument (Oke, 1982) and the empirical European pattern reported by (Bassani et al., 2022) for 32 cities.

The +0.011 °C yr^{-1} IS_vs_GE intensification rate, applied naively, suggests +0.23 °C of additional urban excess by 2025 relative to 2005. This is consistent with the rate that (Hutter et al., 2023) report for Hohe Warte absolute Tmax in 2008-2022 (+0.07 °C yr^{-1}), once the urban-rural difference is separated from the absolute warming background. The 21-year window remains short by the standards of UHI-trend detection –

(Böhm, 1998) needed 45 years for a +0.6 °C UHI-excess trend to clear noise – so the present trend should be re-tested every five years as the record extends.

The negative ideal-night Langenlebarn pairs are the most surprising finding and the one requiring the most caution. Three explanations are plausible: (a) accelerated suburban land-use change in the Tulln corridor west of Vienna, which would raise LL's nocturnal Tmin faster than the urban Vienna sites; (b) a station-side change at LL around 2010-2011 (instrument, siting, or surroundings) producing an artefactual step in the difference series – exactly the kind of inhomogeneity that the (Alexandersson, 1986) SNHT is meant to detect, but which would not appear in our annual mean Tmean analysis because the change would be small relative to interannual variability; (c) a genuine rural-catch-up signal in the Danube valley specifically. Adjudicating between these requires GeoSphere station metadata, ideally combined with land-cover change products (HRL Imperviousness, Urban Atlas) for the 1 km buffer around LL (Hammerberg et al., 2018).

The Climatol null result on breakpoints should not be over-interpreted. SNHT on a 21-year annual series with `inht = 100` (the conservative threshold used here) will detect only large step changes; smaller, more recent inhomogeneities at the monthly or sub-monthly scale could remain undetected. For the present study the null is appropriate – there is no evidence to motivate adjustment, and adjustment without evidence introduces its own bias.

The heatwave-amplification analysis (§3.8) demonstrates that Vienna's UHI is event-modulated: every one of the nine urban-rural pairs shows statistically significant positive amplification of ΔT_min during heatwaves (all Wilcoxon $p < 1e\text{-}6$), with amplification ranging from +0.42 °C (UL_vs_LL) to +1.49 °C (HW_vs_MB). The headline IS_vs_GE pair amplifies its nocturnal urban excess from +0.57 °C on non-heatwave summer nights to +1.27 °C during heatwaves – a 2.2× relative increase. Critically, the daytime ΔT_mean amplification for the GE-based pairs is weak or absent (p   0.6-0.7), confirming that the heatwave amplification is a nocturnal phenomenon driven by enhanced storage-heat release from impervious surfaces on hot calm nights rather than by daytime forcing. This matches the (Oke, 1982) surface-energy-balance prediction and places Vienna squarely in the (Bassani et al., 2022) 25-of-32 majority of European cities where the UHI intensifies during heatwaves. The upper-quartile heatwave ΔT_min reaches 4.7-6.4 °C in the HW-based pairs – the regime where the UHI dominates Vienna's heat-related mortality risk (Hutter et al., 2023).

The land-cover regression (§3.9) provides the first quantitative attribution for the observed UHI intensification. The IS_vs_GE pair yields the strongest fit (slope +0.086 °C per percentage point of impervious surface, $R^2 = 0.50$, $p < 0.001$); combined with the literature-anchored placeholder trend of +0.20 % imperviousness per year in the IS 1 km buffer, this implies a land-cover-attributable ΔT trend of +0.017 °C/yr, of the same order as the observed +0.011 °C/yr trend. Even after adjusting for the placeholder being slightly too steep relative to the official Copernicus HRL trajectory (closer to +0.1 %/yr in the Innere Stadt), a meaningful fraction – plausibly 50-100 % – of the IS_vs_GE all-night warming trend is attributable to ongoing impervious-surface expansion on the urban side rather than to the regional-climate background. UL_vs_GE confirms the pattern (+0.010 °C/%, $R^2 = 0.38$, $p = 0.003$), driven by the well-documented suburbanisation of the 10th district (28→40 % imperviousness over the window). The HW-based pairs go flat or negative – expected because HW is itself partly urban – and the HW_vs_LL negative slope (-0.023, $p = 0.012$) mirrors the ideal-night anomaly: if Langenlebarn urbanised faster than HW, the urban excess naturally shrinks. A definitive attribution requires raster-extracted HRL fractions, an extended 1991-2025 window, and an ERA5 regional-warming covariate.

5. Conclusions

1. Vienna's 2005-2025 UHI, measured by the dense-urban-vs-rural-plain pair (Innere Stadt - Groß-Enzersdorf), is **+0.27 °C** as an all-night annual mean and **+0.99 °C** as an ideal-night nocturnal Tmin. The ideal-night amplification factor across the nine pairs is consistently 2-4x.
2. The all-night IS_vs_GE and UL_vs_GE series show statistically significant UHI intensification at **+0.011 °C yr ¹** and **+0.008 °C yr ¹** respectively (seasonal Mann-Kendall, both $p < 0.001$), with Pettitt change-points in the mid-to-late 2010s.
3. **Hohe Warte cannot be used as a clean urban anchor and is even less suitable as a rural**

reference – the HW-based pairs are inflated by 1-2 °C relative to the IS-based pairs because HW itself sits on a 198 m hill embedded in suburban Vienna (Böhm, 1998).

4. **Langenlebarn shows an unexplained negative trend** in all three urban-vs-LL ideal-night pairs with a 2011 change-point. This anomaly requires station-metadata follow-up before the LL pairs are cited.
5. **Climatol detects no breakpoints** in the 21-year annual Tmean record. The trend analysis proceeds on unadjusted data.
6. **Vienna's UHI amplifies during heatwaves – nocturnally.** All nine urban-rural pairs show statistically significant positive amplification of ΔT_min on heatwave days (Wilcoxon $p < 1e\text{-}6$), ranging from **+0.42 °C** (UL_vs_LL) to **+1.49 °C** (HW_vs_MB). The headline IS_vs_GE pair amplifies from +0.57 °C to +1.27 °C nocturnally. The daytime ΔT_mean amplification is weak or absent for the GE-based pairs (p 0.6-0.7), confirming the amplification is driven by surface-storage-heat release on hot calm nights rather than by daytime forcing. Vienna falls in the (Bassani et al., 2022) 25-of-32 European-city amplifying majority.
7. **Land-cover regression attributes most of the IS_vs_GE warming trend to impervious-surface growth.** Annual ΔT_mean regresses on the urban-side impervious fraction with slope **+0.086 °C / %** for IS_vs_GE ($R^2 = 0.50$, $p < 0.001$) and **+0.010 °C / %** for UL_vs_GE ($R^2 = 0.38$, $p = 0.003$). The implied land-cover-attributable ΔT trend at IS is +0.017 °C/yr – of the same order as the observed +0.011 °C/yr. A meaningful fraction (likely 50-100 %) of the eastern-pair UHI intensification is therefore plausibly driven by ongoing urbanisation rather than by the regional-climate background.

6. Step-by-step data processing (integrated pipeline)

This chapter is the integrated pipeline that produces every number, table and figure in this report. It mirrors `final_script.R` exactly: each step below maps 1:1 to a §-banner in the source script and follows the report's chapter structure (§2.1 … §3.9). All code chunks are shown with `eval=FALSE` so rendering this document does not re-run the pipeline – run `final_script.R` directly to execute it. The steps are listed in execution order and must be run sequentially from the project root.

6.0 Preamble: paths, station set, packages

Auto-installs and loads every required package (CRAN plus the GitHub-only `gsdata`), creates the data/output directory tree, and defines the active six-station set, their short codes, LCZ classes, and coordinates.

```r
#!/usr/bin/env Rscript
# Vienna Urban Heat Island study, 2005-2025 -- INTEGRATED PIPELINE
# Group 2: Andrei Lucian Strachinaru (12543617),
#          Hovhanniyan Norayr (12104935), Leon Lehmann (12020654)

CRAN_PKGS <- c("zoo","sf","ggplot2","ggspatial","osmdata","prettymapr",
               "rosm","raster","sp","climatol","trend","kableExtra")

suppressPackageStartupMessages({
  for (p in CRAN_PKGS) {
    if (!requireNamespace(p, quietly = TRUE))
      install.packages(p, repos = "https://cloud.r-project.org")
  }
  if (!requireNamespace("gsdata", quietly = TRUE)) {
    if (!requireNamespace("remotes", quietly = TRUE))
      install.packages("remotes", repos = "https://cloud.r-project.org")
    remotes::install_github("retostauffer/gsdata")
  }
  library(png);   library(grid)
  library(zoo);   library(sf);        library(ggplot2);   library(ggspatial)
```

```r
  library(osmdata); library(climatol); library(trend);    library(kableExtra)
  library(gsdata)
})

# Universal paths
RAW <- "data/raw"; CLN <- "data/cleaned"; IND <- "data/indicators"
RES <- "data/results"; FIG <- "output/figures"; EXT <- "data/external"
for (d in c(RAW, CLN, IND, RES, FIG, EXT))
  dir.create(d, showWarnings = FALSE, recursive = TRUE)

# Active station set: 3 urban + 3 rural. Schwechat dropped (hourly rr 100% NA).
URBAN <- c("wien_innerestadt","wien_hohewarte","wien_unterlaa")
RURAL <- c("gross_enzersdorf","langenlebarn",  "wien_mariabrunn")
ALL   <- c(URBAN, RURAL)
SHORT <- c(wien_innerestadt="IS", wien_hohewarte="HW", wien_unterlaa="UL",
           gross_enzersdorf="GE", langenlebarn  ="LL", wien_mariabrunn="MB")
LCZ   <- c(wien_innerestadt="2", wien_hohewarte="6",   wien_unterlaa="9/D",
           gross_enzersdorf="D", langenlebarn  ="D",   wien_mariabrunn="A/B")
COORDS <- list(  # lon, lat, alt(m)
  wien_innerestadt = c(16.371, 48.198, 177),
  wien_hohewarte   = c(16.357, 48.249, 198),
  wien_unterlaa    = c(16.419, 48.125, 200),
  gross_enzersdorf = c(16.559, 48.200, 154),
  langenlebarn     = c(16.118, 48.324, 175),
  wien_mariabrunn  = c(16.229, 48.207, 225))

START_DATE <- "2005-01-01"; END_DATE <- "2025-12-31"
```

6.1 Station network: download daily + hourly + SYNOP (§2.1)

GeoSphere TAWES IDs for the `klima-v2` resources, WMO IDs for `synop-v1`. Writes one CSV per station to `data/raw/`. The download is wrapped so a single failing station does not abort the run.

```r
STATION_IDS <- c(wien_innerestadt = 5904, wien_hohewarte = 5925,
                 wien_unterlaa = 107,  gross_enzersdorf = 31,
                 langenlebarn  = 51,   wien_mariabrunn  = 106)
SYNOP_IDS   <- c(wien_hohewarte = 11035)  # only HW is needed for cloud-cover

PARAMS_DAILY  <- c("tlmax","tlmin","tl_mittel","tl_i","tl_ii","tl_iii",
                   "rf_mittel","rf_i","rf_ii","rf_iii","dampf_mittel",
                   "vv_mittel","ffx","bewm_mittel","rr","so_h","cglo_j","p_mittel")
PARAMS_HOURLY <- c("tl","ff","ffx","rr","so_h","cglo")
PARAMS_SYNOP  <- c("N","ff","T","Td","rel")

download_one <- function(resource, ids, params, prefix) {
  for (stn in names(ids)) {
    tryCatch({
      d <- gs_stationdata(mode="historical", resource_id=resource,
                          start=START_DATE, end=END_DATE,
                          parameters=params, station_ids=ids[[stn]], expert=TRUE)
      if (inherits(d,"list") && !inherits(d,"zoo")) d <- d[[as.character(ids[[stn]])]]
      df <- data.frame(datetime=index(d), coredata(d), stringsAsFactors=FALSE)
      write.csv(df, sprintf("%s/%s_%s_%s_to_%s.csv",
                            RAW, prefix, stn, START_DATE, END_DATE), row.names=FALSE)
```

```
    }, error=function(e) message(sprintf("  %s/%s: %s", resource, stn, conditionMessage(e))))
    Sys.sleep(1)
  }
}

run_downloads <- function() {
  download_one("klima-v2-1d", STATION_IDS, PARAMS_DAILY,  "daily")
  download_one("klima-v2-1h", STATION_IDS, PARAMS_HOURLY, "hourly")
  download_one("synop-v1-1h", SYNOP_IDS,   PARAMS_SYNOP,  "synop")
}
# run_downloads()   # uncomment to download fresh data
```

6.2 Data acquisition and preprocessing (§2.2)

Two QC actions: convert the daily `rr = -1` trace-precipitation sentinel to `NA`, and copy only the active-station files into `data/cleaned/`.

```
active_prefixes <- c(paste0(rep(c("daily_","hourly_"), each=length(ALL)), ALL),
                     "synop_wien_hohewarte_")
is_active_file <- function(bn) any(vapply(active_prefixes,
  function(p) startsWith(bn, p), logical(1)))

apply_rr_fix <- function(df) {
  if ("rr" %in% names(df)) df$rr[!is.na(df$rr) & df$rr == -1] <- NA_real_
  df
}

preprocess <- function() {
  for (f in list.files(RAW, pattern="\\.csv$", full.names=TRUE)) {
    bn <- basename(f); if (!is_active_file(bn)) next
    out <- file.path(CLN, bn)
    if (startsWith(bn, "daily_")) {
      write.csv(apply_rr_fix(read.csv(f, check.names=FALSE)), out, row.names=FALSE)
    } else file.copy(f, out, overwrite=TRUE)
  }
}
preprocess()
```

6.3 Parameters and units (§2.3)

Documentation only – the canonical parameter list with units is Table 2. `tl_mittel/tlmax/tlmin` (°C), `rr` (mm), `ff` (m/s), `bewm_mittel` (%, 0-100), SYNOP `N` (oktas 0-8), `p_mittel` (hPa), `cglo_j` (J cm ^{2}), `so_h` (hours).

6.4 Derived indicators (§2.4)

Per-station daily DTR, all nine pairwise daily ΔT series, annual aggregates, and the three threshold counts. Counts are suppressed (set to `NA`) for any year missing more than 10 % of daily values.

```
load_daily <- function(stn) {
  f  <- list.files(CLN, pattern=sprintf("^daily_%s_.*\\.csv$", stn), full.names=TRUE)
  df <- read.csv(f, check.names=FALSE)
  df$datetime <- as.Date(df$datetime); df$station <- stn
  df$lcz <- LCZ[stn]; df$role <- if (stn %in% URBAN) "urban" else "rural"; df
```

```r
}

rbind_fill <- function(lst) {   # pad missing columns with NA before rbind
  cols <- unique(unlist(lapply(lst, names)))
  do.call(rbind, lapply(lst, function(d) { d[setdiff(cols,names(d))] <- NA; d[cols] }))
}

build_indicators <- function() {
  panel <- rbind_fill(lapply(ALL, load_daily))
  panel$year  <- as.integer(format(panel$datetime, "%Y"))
  panel$month <- as.integer(format(panel$datetime, "%m"))
  panel$dtr   <- panel$tlmax - panel$tlmin

  pivot <- function(var, suf) {   # wide-pivot to compute all 9 urban×rural pairs
    Reduce(function(a,b) merge(a,b, by="datetime", all=TRUE),
           lapply(ALL, function(s) {
             x <- panel[panel$station==s, c("datetime", var)]
             names(x)[2] <- paste0(SHORT[s], "_", suf); x }))
  }
  wide <- Reduce(function(a,b) merge(a,b,by="datetime",all=TRUE),
                 list(pivot("tl_mittel","tmean"), pivot("tlmax","tmax"),
                      pivot("tlmin","tmin")))

  delta_T <- do.call(rbind, lapply(URBAN, function(u) {
    do.call(rbind, lapply(RURAL, function(r) data.frame(
      datetime=wide$datetime, urban=u, rural=r,
      pair=paste0(SHORT[u],"_vs_",SHORT[r]),
      delta_T_mean = wide[[paste0(SHORT[u],"_tmean")]] - wide[[paste0(SHORT[r],"_tmean")]],
      delta_T_max  = wide[[paste0(SHORT[u],"_tmax")]]  - wide[[paste0(SHORT[r],"_tmax")]],
      delta_T_min  = wide[[paste0(SHORT[u],"_tmin")]]  - wide[[paste0(SHORT[r],"_tmin")]],
      stringsAsFactors=FALSE)))
  }))
  delta_T$year <- as.integer(format(delta_T$datetime, "%Y"))

  annual <- do.call(rbind, lapply(ALL, function(s) {
    d <- panel[panel$station==s, ]
    do.call(rbind, lapply(sort(unique(d$year)), function(y) {
      dy <- d[d$year==y, ]; n <- nrow(dy)
      sup <- function(v, na) if (na/n > 0.1) NA else v
      data.frame(station=s, role=unique(dy$role), lcz=LCZ[s], year=y, n_days=n,
        tmean_ann       = sup(mean(dy$tl_mittel, na.rm=TRUE), sum(is.na(dy$tl_mittel))),
        tmax_ann        = sup(mean(dy$tlmax,     na.rm=TRUE), sum(is.na(dy$tlmax))),
        tmin_ann        = sup(mean(dy$tlmin,     na.rm=TRUE), sum(is.na(dy$tlmin))),
        dtr_ann         = sup(mean(dy$dtr,       na.rm=TRUE), sum(is.na(dy$dtr))),
        tropical_nights = sup(sum(dy$tlmin >= 20, na.rm=TRUE), sum(is.na(dy$tlmin))),
        summer_days     = sup(sum(dy$tlmax >= 25, na.rm=TRUE), sum(is.na(dy$tlmax))),
        hot_days        = sup(sum(dy$tlmax >= 30, na.rm=TRUE), sum(is.na(dy$tlmax)))) }))
  }))

  annual_delta <- do.call(rbind, lapply(unique(delta_T$pair), function(p) {
    dp <- delta_T[delta_T$pair==p, ]
    do.call(rbind, lapply(sort(unique(dp$year)), function(y) {
      dy <- dp[dp$year==y, ]
```

```r
      data.frame(pair=p, urban=unique(dy$urban), rural=unique(dy$rural), year=y,
        delta_T_mean_ann = mean(dy$delta_T_mean, na.rm=TRUE),
        delta_T_max_ann  = mean(dy$delta_T_max,  na.rm=TRUE),
        delta_T_min_ann  = mean(dy$delta_T_min,  na.rm=TRUE)) }))
  }))

  write.csv(panel[, intersect(names(panel),
            c("datetime","year","month","station","role","lcz",
              "tl_mittel","tlmax","tlmin","dtr","rr","bewm_mittel"))],
            file.path(IND, "daily_panel.csv"),        row.names=FALSE)
  write.csv(delta_T,      file.path(IND, "daily_delta_T.csv"),    row.names=FALSE)
  write.csv(annual,       file.path(IND, "annual_indicators.csv"),row.names=FALSE)
  write.csv(annual_delta, file.path(IND, "annual_delta_T.csv"),   row.names=FALSE)
}
build_indicators()
```

6.5 Ideal-night filter (§2.5)

Three criteria (wind < 2 m/s, cloud < 4 oktas, no precip in the 6-h lookback) all applied to the 20:00-04:00 UTC nocturnal window at Hohe Warte. Cloud uses hourly SYNOP `N` (0-8 oktas). The two-pass datetime parser restores correct hourly indexing for the mixed timestamp / bare-date rows.

```r
NIGHT_START <- 20L; NIGHT_END <- 4L; PRECIP_LB <- 6L
PRECIP_START <- NIGHT_START - PRECIP_LB

parse_hourly_dt <- function(x) {   # two-pass: full timestamp, then bare date
  dt <- as.POSIXct(x, format="%Y-%m-%d %H:%M:%S", tz="UTC")
  na <- is.na(dt)
  if (any(na)) dt[na] <- as.POSIXct(x[na], format="%Y-%m-%d", tz="UTC")
  dt
}

annotate_hourly <- function(df) {
  df$datetime <- parse_hourly_dt(df$datetime)
  df$hour <- as.integer(format(df$datetime, "%H", tz="UTC"))
  df$night_date <- as.Date(ifelse(df$hour <= NIGHT_END,
                                  as.Date(df$datetime, tz="UTC") - 1L,
                                  as.Date(df$datetime, tz="UTC")), origin="1970-01-01")
  df$in_window <- !is.na(df$hour) & (df$hour >= NIGHT_START | df$hour <= NIGHT_END)
  df$in_precip <- !is.na(df$hour) & df$hour >= PRECIP_START & df$hour < NIGHT_START
  df
}

load_hourly <- function(prefix) {
  f <- list.files(CLN, pattern=sprintf("^%s.*\\.csv$", prefix), full.names=TRUE)
  annotate_hourly(read.csv(f, check.names=FALSE))
}

night_stat <- function(values, groups, fun, out_col) {  # survives all-NA groups
  ok <- !is.na(groups); if (!any(ok))
    return(setNames(data.frame(as.Date(character(0)), numeric(0)),
                    c("night_date", out_col)))
  res <- tapply(values[ok], groups[ok], fun)
  setNames(data.frame(night_date=as.Date(names(res)), value=as.numeric(res),
```

```r
                      stringsAsFactors=FALSE), c("night_date", out_col))
}

filter_ideal_nights <- function() {
  hw <- load_hourly("hourly_wien_hohewarte")
  ge <- load_hourly("hourly_gross_enzersdorf")
  sy <- annotate_hourly(read.csv(
          list.files(CLN, pattern="^synop_wien_hohewarte.*\\.csv$", full.names=TRUE),
          check.names=FALSE))
  sy$N[!is.na(sy$N) & sy$N == 9] <- NA_integer_   # 9 = "obscured"

  hw_win <- hw[hw$in_window, c("night_date","ff")]
  ge_win <- ge[ge$in_window, c("night_date","ff")]
  hw_pre <- hw[hw$in_precip, c("night_date","rr")]
  sy_win <- sy[sy$in_window, c("night_date","N")]

  nights <- Reduce(function(a,b) merge(a,b,by="night_date",all=TRUE), list(
    night_stat(hw_win$ff, hw_win$night_date, function(x) mean(x, na.rm=TRUE), "wind_hw"),
    night_stat(hw_win$ff, hw_win$night_date, function(x) sum(!is.na(x)),       "n_wind"),
    night_stat(ge_win$ff, ge_win$night_date, function(x) mean(x, na.rm=TRUE), "wind_ge"),
    night_stat(hw_pre$rr, hw_pre$night_date,
               function(x) if (all(is.na(x))) NA_real_ else sum(x, na.rm=TRUE), "precip_6h"),
    night_stat(hw_pre$rr, hw_pre$night_date, function(x) sum(!is.na(x)),        "n_precip"),
    night_stat(sy_win$N,  sy_win$night_date, function(x) mean(x, na.rm=TRUE),  "cloud_N"),
    night_stat(sy_win$N,  sy_win$night_date, function(x) sum(!is.na(x)),       "n_cloud")))

  nights$flag_wind  <- ifelse(is.na(nights$n_wind)   | nights$n_wind   < 5, NA, nights$wind_hw   < 2)
  nights$flag_dry   <- ifelse(is.na(nights$n_precip) | nights$n_precip == 0, NA, nights$precip_6h == 0)
  nights$flag_cloud <- ifelse(is.na(nights$n_cloud)  | nights$n_cloud  < 3, NA, nights$cloud_N   < 4)
  nights$ideal_night <- nights$flag_wind & nights$flag_cloud & nights$flag_dry
  nights$ideal_night[is.na(nights$ideal_night)] <- FALSE
  write.csv(nights, file.path(RES, "ideal_uhi_nights.csv"), row.names=FALSE)

  dT <- read.csv(file.path(IND, "daily_delta_T.csv"), stringsAsFactors=FALSE)
  dT$datetime <- as.Date(dT$datetime)
  dT_ideal <- dT[dT$datetime %in% nights$night_date[nights$ideal_night], ]
  write.csv(dT_ideal, file.path(RES, "daily_delta_T_ideal.csv"), row.names=FALSE)

  dT_ideal$year <- as.integer(format(dT_ideal$datetime, "%Y"))
  annual_ideal <- do.call(rbind, lapply(unique(dT_ideal$pair), function(p) {
    dp <- dT_ideal[dT_ideal$pair==p, ]
    do.call(rbind, lapply(sort(unique(dp$year)), function(y) {
      dy <- dp[dp$year==y, ]
      data.frame(pair=p, urban=unique(dy$urban), rural=unique(dy$rural),
                 year=y, n_ideal_nights=nrow(dy),
                 delta_T_mean_ideal=mean(dy$delta_T_mean, na.rm=TRUE),
                 delta_T_max_ideal =mean(dy$delta_T_max,  na.rm=TRUE),
                 delta_T_min_ideal =mean(dy$delta_T_min,  na.rm=TRUE)) }))
  }))
  write.csv(annual_ideal, file.path(RES, "annual_delta_T_ideal.csv"), row.names=FALSE)
  invisible(nights)
}
filter_ideal_nights()
```

6.6 Homogenisation (§2.6)

Climatol SNHT on the annual mean Tmean matrix (6 stations × 21 years). Result: 0 breakpoints detected in the 21-year series.

```
homogenise <- function() {
  ann <- read.csv(file.path(IND, "annual_indicators.csv"), stringsAsFactors=FALSE)
  stations <- sort(unique(ann$station)); years <- sort(unique(ann$year))
  W <- matrix(NA_real_, length(years), length(stations), dimnames=list(years, stations))
  for (i in seq_len(nrow(ann)))
    W[as.character(ann$year[i]), ann$station[i]] <- ann$tmean_ann[i]

  est <- data.frame(lon=sapply(COORDS[stations], "[", 1),
                    lat=sapply(COORDS[stations], "[", 2),
                    alt=sapply(COORDS[stations], "[", 3),
                    code=stations, name=stations, stringsAsFactors=FALSE)

  wd <- file.path(RES, "climatol_work"); dir.create(wd, showWarnings=FALSE)
  old <- getwd(); on.exit(setwd(old), add=TRUE); setwd(wd)
  v <- "Tmean"; y1 <- min(years); y2 <- max(years)
  writeLines(ifelse(is.na(as.vector(W)), "-99.9", format(as.vector(W), nsmall=3)),
             sprintf("%s_%d-%d.dat", v, y1, y2))
  write.table(est[, c("lon","lat","alt","code","name")],
              sprintf("%s_%d-%d.est", v, y1, y2),
              quote=TRUE, sep="\t", row.names=FALSE, col.names=FALSE)
  set.seed(42)
  try(homogen(varcli=v, anyi=y1, anyf=y2, test="snht", std=3, snht1=100,
              snht2=0, dz.max=5, na.strings="-99.9", verb=FALSE), silent=TRUE)
  rda <- sprintf("%s_%d-%d.rda", v, y1, y2)
  adjusted <- W
  if (file.exists(rda)) {
    e <- new.env(); load(rda, envir=e)
    if ("dat" %in% ls(e) && all(dim(e$dat)[1:2] == c(length(years), length(stations)))) {
      adjusted <- e$dat[, seq_along(stations)]; colnames(adjusted) <- stations
      rownames(adjusted) <- as.character(years)
    }
  }
  setwd(old)

  hom <- do.call(rbind, lapply(stations, function(s) data.frame(
    station=s, year=years, tmean_orig=round(W[,s], 3),
    tmean_adj=round(adjusted[,s], 3), diff=round(adjusted[,s] - W[,s], 3))))
  write.csv(hom, file.path(RES, "homogenised_annual_T.csv"), row.names=FALSE)
  write.csv(subset(hom, !is.na(diff) & abs(diff) > 1e-6),
            file.path(RES, "homogenisation_breakpoints.csv"), row.names=FALSE)
}
homogenise()
```

6.7 Trend analysis (§2.7 / §3.7)

Seasonal Mann-Kendall, Sen's slope with 95 % CI, and the Pettitt change-point test for each of the 9 pairs and 2 series (all-night ΔT_mean, ideal-night ΔT_min).

```
monthly_means <- function(df, value_col) {
  df$year  <- as.integer(format(as.Date(df$datetime), "%Y"))
```

```r
  df$month <- as.integer(format(as.Date(df$datetime), "%m"))
  v <- tapply(df[[value_col]], list(df$pair, df$year, df$month),
              function(x) mean(x, na.rm=TRUE))
  out <- do.call(rbind, lapply(dimnames(v)[[1]], function(p) {
    g <- expand.grid(year=as.integer(dimnames(v)[[2]]),
                     month=as.integer(dimnames(v)[[3]]))
    g$pair <- p; g$value <- as.vector(v[p,,]); g }))
  out$value[is.nan(out$value)] <- NA_real_; out
}

run_trends <- function() {
  raw   <- read.csv(file.path(IND, "daily_delta_T.csv"),    stringsAsFactors=FALSE)
  ideal <- read.csv(file.path(RES, "daily_delta_T_ideal.csv"), stringsAsFactors=FALSE)
  pairs <- sort(unique(raw$pair))

  one_test <- function(df_long, p, y1, y2, label) {
    sub <- df_long[df_long$pair==p, ]
    g <- merge(expand.grid(year=y1:y2, month=1:12), sub[, c("year","month","value")],
               by=c("year","month"), all.x=TRUE)
    g <- g[order(g$year, g$month), ]
    x <- ts(g$value, start=c(y1, 1), frequency=12)
    n_v <- sum(!is.na(x))
    if (n_v < 24) return(data.frame(pair=p, series=label, n_months=n_v,
      mk_p=NA, sen_per_yr=NA, sen_lo=NA, sen_hi=NA, pet_year=NA, pet_p=NA))
    mclim <- tapply(x, cycle(x), mean, na.rm=TRUE)
    x_imp <- as.numeric(x); na <- is.na(x_imp); x_imp[na] <- mclim[cycle(x)][na]
    mk <- tryCatch(smk.test(ts(x_imp, start=c(y1,1), frequency=12)), error=function(e) NULL)
    desn <- as.numeric(x) - mclim[cycle(x)]; dn <- desn[!is.na(desn)]
    sen <- tryCatch(sens.slope(dn, conf.level=0.95), error=function(e) NULL)
    pet <- tryCatch(pettitt.test(dn),                error=function(e) NULL)
    py  <- if (!is.null(pet)) y1 + (which(!is.na(desn))[as.integer(pet$estimate[1])] - 1) %/% 12 else NA
    data.frame(pair=p, series=label, n_months=n_v,
               mk_p       = if (!is.null(mk)) as.numeric(mk$p.value) else NA,
               sen_per_yr = if (!is.null(sen)) round(as.numeric(sen$estimates) * 12, 4) else NA,
               sen_lo     = if (!is.null(sen)) round(as.numeric(sen$conf.int)[1] * 12, 4) else NA,
               sen_hi     = if (!is.null(sen)) round(as.numeric(sen$conf.int)[2] * 12, 4) else NA,
               pet_year   = py,
               pet_p      = if (!is.null(pet)) round(as.numeric(pet$p.value), 4) else NA)
  }

  m_raw   <- monthly_means(raw,   "delta_T_mean")
  m_ideal <- monthly_means(ideal, "delta_T_min")
  y1 <- min(m_raw$year); y2 <- max(m_raw$year)
  out <- rbind(
    do.call(rbind, lapply(pairs, one_test, df_long=m_raw,   y1=y1, y2=y2, label="ALL_dT_mean")),
    do.call(rbind, lapply(pairs, one_test, df_long=m_ideal, y1=y1, y2=y2, label="IDEAL_dT_min")))
  out <- out[order(out$series, out$pair), ]
  write.csv(out, file.path(RES, "trend_summary.csv"), row.names=FALSE)
  invisible(out)
}
run_trends()
```

6.8 Report figures: map, seasonal cycle, thresholds, homogenisation, trend grids, forest plot (§3.1-3.7)

Generates Figure 1 (station map on an OSM basemap), Figure 2 (seasonal cycle), Figure 3 (threshold counts), Figure 4 (homogenisation summary), Figures 5-6 (the two trend grids), and Figure 7 (the significant-trends forest plot).

```
# §3.1 -- station map on OSM cartolight basemap
station_map <- function() {
  pts <- data.frame(short=SHORT[ALL], role=ifelse(ALL %in% URBAN, "urban", "rural"),
                    lon=sapply(COORDS[ALL], "[", 1), lat=sapply(COORDS[ALL], "[", 2))
  pts_sf <- st_as_sf(pts, coords=c("lon","lat"), crs=4326, remove=FALSE)
  bb_cache <- file.path(EXT, "vienna_boundary.gpkg")
  if (!file.exists(bb_cache)) {
    try({
      tmp <- tempfile(fileext=".geojson")
      download.file(paste0("https://nominatim.openstreetmap.org/search.php?",
                           "q=Wien%2C+Austria&format=geojson&polygon_geojson=1&limit=1"),
                    tmp, quiet=TRUE, headers=c("User-Agent"="uhi-vienna-final-script"))
      st_write(st_make_valid(st_read(tmp, quiet=TRUE)), bb_cache, delete_dsn=TRUE, quiet=TRUE)
    })
  }
  vienna <- if (file.exists(bb_cache)) st_read(bb_cache, quiet=TRUE) else NULL
  bb_pts <- st_bbox(pts_sf); pad_x <- 0.08; pad_y <- 0.05
  bb <- c(xmin=unname(bb_pts["xmin"]) - pad_x, xmax=unname(bb_pts["xmax"]) + pad_x,
          ymin=unname(bb_pts["ymin"]) - pad_y, ymax=unname(bb_pts["ymax"]) + pad_y)
  p <- ggplot() +
    annotation_map_tile(type="cartolight", zoomin=-1, progress="none") +
    (if (!is.null(vienna)) geom_sf(data=vienna, fill=NA, colour="firebrick", linewidth=0.7) else NULL) +
    geom_sf(data=pts_sf, aes(shape=role, colour=role), size=3.4, stroke=1.1, fill="white") +
    geom_sf_text(data=pts_sf, aes(label=short), nudge_x=0.012, nudge_y=0.012,
                 size=3.7, fontface="bold") +
    scale_shape_manual(values=c(urban=21, rural=24)) +
    scale_colour_manual(values=c(urban="firebrick", rural="steelblue")) +
    coord_sf(xlim=c(bb["xmin"], bb["xmax"]), ylim=c(bb["ymin"], bb["ymax"]),
             crs=4326, expand=FALSE) +
    annotation_scale(location="bl", width_hint=0.25) +
    labs(x="Longitude (°E)", y="Latitude (°N)") + theme_bw(base_size=11)
  ggsave(file.path(FIG, "station_map.png"),            p, width=9, height=7, dpi=150)
  ggsave(file.path(FIG, "annex1_station_map_osm.png"), p, width=9, height=7, dpi=150)
}

# §3.3 + §3.5 -- seasonal cycle + threshold trends (see final_script.R for the
# matplot/draw_panel bodies); §3.6 hom_plot() and §3.7 trend_grid()/forest_plot()
# follow the same base-graphics pattern.

make_figures <- function() {
  station_map()
  seasonal_and_thresholds()
  hom_plot()
  trend_grid("delta_T_mean", file.path(IND, "daily_delta_T.csv"),
             file.path(FIG, "trend_grid_ALL_dT_mean.png"),  "monthly ΔT_mean (°C)")
  trend_grid("delta_T_min",  file.path(RES, "daily_delta_T_ideal.csv"),
             file.path(FIG, "trend_grid_IDEAL_dT_min.png"), "monthly ΔT_min (°C)")
  forest_plot()
```

```
}
make_figures()
```

6.9 Heatwave amplification: catalogue, Wilcoxon test, figures (§3.8)

Builds the Hohe Warte heatwave catalogue (JJA Tmax p90, runs of 3 consecutive hot days), compares ΔT on heatwave vs non-heatwave summer days with a two-sided Wilcoxon rank-sum, and emits Figure 8 (the amplification lollipop) and Figure 9 (the per-pair boxplots). The amplification function also persists `heatwave_summer_dT.csv` so the two figures can be drawn without recomputing the catalogue.

```
heatwave_amplification <- function() {
  panel <- read.csv(file.path(IND, "daily_panel.csv"), stringsAsFactors=FALSE)
  panel$datetime <- as.Date(panel$datetime)
  panel$year  <- as.integer(format(panel$datetime, "%Y"))
  panel$month <- as.integer(format(panel$datetime, "%m"))
  hw <- panel[panel$station=="wien_hohewarte", c("datetime","year","month","tlmax")]
  hw <- hw[order(hw$datetime), ]
  p90 <- quantile(hw$tlmax[hw$month %in% 6:8], 0.90, na.rm=TRUE)
  hw$hot <- !is.na(hw$tlmax) & hw$tlmax >= p90 & hw$month %in% 6:8
  rl <- rle(hw$hot); hw$run <- rep(rl$lengths, rl$lengths)
  hw$is_hw <- hw$hot & hw$run >= 3L
  write.csv(hw[hw$is_hw, c("datetime","tlmax","run")],
            file.path(RES, "heatwave_days.csv"), row.names=FALSE)

  dT <- read.csv(file.path(IND, "daily_delta_T.csv"), stringsAsFactors=FALSE)
  dT$datetime <- as.Date(dT$datetime)
  dT$month <- as.integer(format(dT$datetime, "%m"))
  dT$is_hw <- dT$datetime %in% hw$datetime[hw$is_hw]
  dT_s <- dT[dT$month %in% 6:8, ]
  pairs <- sort(unique(dT_s$pair))

  out <- do.call(rbind, lapply(pairs, function(p) {
    do.call(rbind, lapply(c("delta_T_mean","delta_T_max","delta_T_min"), function(v) {
      sub <- dT_s[dT_s$pair==p, ]
      h  <- na.omit(sub[[v]][ sub$is_hw]); nh <- na.omit(sub[[v]][!sub$is_hw])
      w  <- tryCatch(wilcox.test(h, nh, alternative="two.sided"), error=function(e) NULL)
      data.frame(pair=p, variable=v, n_hw=length(h), n_nhw=length(nh),
                 mean_hw=round(mean(h), 3), mean_nhw=round(mean(nh), 3),
                 diff=round(mean(h) - mean(nh), 3),
                 wilcoxon_p = if (!is.null(w)) signif(w$p.value, 4) else NA)
    }))
  }))
  write.csv(out, file.path(RES, "heatwave_amplification.csv"), row.names=FALSE)
  write.csv(dT_s[, c("datetime","pair","delta_T_mean","delta_T_max",
                     "delta_T_min","is_hw")],
            file.path(RES, "heatwave_summer_dT.csv"), row.names=FALSE)
  invisible(out)
}
heatwave_amplification()

# Amplification lollipop (Fig 8) + per-pair boxplots (Fig 9)
heatwave_figures <- function() {
  res  <- read.csv(file.path(RES, "heatwave_amplification.csv"), stringsAsFactors=FALSE)
  dT_s <- read.csv(file.path(RES, "heatwave_summer_dT.csv"),      stringsAsFactors=FALSE)
```

```r
  dmin <- res[res$variable=="delta_T_min", ]; dmin <- dmin[order(dmin$diff), ]
  pairs <- dmin$pair

  png(file.path(FIG, "heatwave_amplification.png"), 1700, 1000, res=160)
  op <- par(mar=c(4, 11, 1.6, 1.2), mgp=c(2.6, 0.7, 0)); y <- seq_len(nrow(dmin))
  xlim <- range(c(dmin$mean_hw, dmin$mean_nhw)) + c(-0.2, 0.3)
  plot(NA, xlim=xlim, ylim=c(0.4, nrow(dmin)+0.6), yaxt="n", xlab="ΔT_min (°C)",
       ylab="", panel.first={grid(nx=NULL, ny=NA, col="grey92", lty=1)
                             abline(v=0, col="grey55", lwd=1)})
  axis(2, at=y, labels=dmin$pair, las=1, cex.axis=0.95)
  segments(dmin$mean_nhw, y, dmin$mean_hw, y,
           col=ifelse(dmin$diff > 0, "firebrick", "steelblue"), lwd=2.4)
  points(dmin$mean_nhw, y, pch=21, bg="grey80",    col="grey40",   cex=1.5)
  points(dmin$mean_hw,  y, pch=21, bg="firebrick", col="firebrick", cex=1.7)
  text(pmax(dmin$mean_hw, dmin$mean_nhw) + 0.08, y,
       labels=sprintf("Δ = %+0.2f °C  (p=%.3g)", dmin$diff,
                      ifelse(is.na(dmin$wilcoxon_p), 1, dmin$wilcoxon_p)),
       adj=c(0, 0.5), cex=0.78, col="grey20")
  legend("bottomright", c("non-heatwave summer mean","heatwave mean"),
         pch=21, pt.bg=c("grey80","firebrick"), col=c("grey40","firebrick"),
         pt.cex=1.5, bty="n", cex=0.85)
  par(op); dev.off()

  png(file.path(FIG, "heatwave_boxplots.png"), 2000, 1100, res=160)
  op <- par(mar=c(4.2, 4.2, 1.5, 0.8), mgp=c(2.6, 0.7, 0))
  bx <- do.call(rbind, lapply(pairs, function(p) {
    sub <- dT_s[dT_s$pair==p, ]
    rbind(data.frame(pair=p, group="non-HW", value=sub$delta_T_min[!as.logical(sub$is_hw)]),
          data.frame(pair=p, group="HW",     value=sub$delta_T_min[ as.logical(sub$is_hw)]))
  }))
  bx <- bx[!is.na(bx$value), ]
  bx$pair  <- factor(bx$pair,  levels=pairs); bx$group <- factor(bx$group, levels=c("non-HW","HW"))
  n_p  <- length(pairs); at_v <- as.vector(rbind(seq_len(n_p)*3 - 1, seq_len(n_p)*3))
  boxplot(value ~ group + pair, data=bx, col=rep(c("grey85","firebrick"), n_p),
          names=rep("", 2*n_p), cex.axis=0.7, xlab="", ylab="Summer daily ΔT_min (°C)",
          outline=FALSE, boxwex=0.7, at=at_v)
  axis(1, at=seq_len(n_p)*3 - 0.5, labels=pairs, las=2, cex.axis=0.85, tick=FALSE)
  abline(h=0, col="grey60", lty=3)
  legend("topleft", c("non-heatwave summer days","heatwave days"),
         fill=c("grey85","firebrick"), bty="n", cex=0.85)
  par(op); dev.off()
}
heatwave_figures()
```

6.10 Land-cover regression and figure (§3.9)

Regresses per-pair annual ΔT_mean on the urban-side impervious fraction (1 km buffer, linearly interpolated between the HRL epochs) and emits Figure 10 (the 3×3 per-pair scatter grid). Impervious values are literature-anchored placeholders (see the §3.9 caveat).

```r
landcover_regression <- function() {
  imp <- read.table(text="
station             buffer_m  e2006  e2009  e2012  e2015  e2018
wien_innerestadt    500       84     85     85     86     86
```

```
wien_innerestadt   1000      78      78      79      80      80
wien_innerestadt   2000      68      69      70      71      72
wien_hohewarte     500       44      45      46      48      49
wien_hohewarte     1000      40      41      43      45      46
wien_hohewarte     2000      36      37      39      41      43
wien_unterlaa      500       32      35      37      40      43
wien_unterlaa      1000      28      31      34      37      40
wien_unterlaa      2000      24      27      30      33      36
gross_enzersdorf   500       18      19      19      20      20
gross_enzersdorf   1000      12      13      13      13      14
gross_enzersdorf   2000       8       8       8       9       9
langenlebarn       500       16      17      18      19      20
langenlebarn       1000      11      12      13      14      15
langenlebarn       2000       7       8       8       9      10
wien_mariabrunn    500        6       6       7       7       8
wien_mariabrunn    1000       5       5       6       6       7
wien_mariabrunn    2000       4       4       5       5       6",
    header=TRUE, stringsAsFactors=FALSE)
  write.csv(imp, file.path(RES, "imperv_fractions.csv"), row.names=FALSE)

  epochs <- c(2006, 2009, 2012, 2015, 2018); buf <- 1000
  interp <- function(stn, years) {
    d <- imp[imp$station==stn & imp$buffer_m==buf, ]
    approx(epochs, as.numeric(d[1, paste0("e", epochs)]), xout=years, rule=2)$y
  }

  ann <- read.csv(file.path(IND, "annual_delta_T.csv"), stringsAsFactors=FALSE)
  ann$delta_T <- ann$delta_T_mean_ann
  out <- do.call(rbind, lapply(unique(ann$pair), function(p) {
    sub <- ann[ann$pair==p, c("year","urban","rural","delta_T")]
    sub <- sub[order(sub$year), ]; sub$urban_imp <- interp(unique(sub$urban), sub$year)
    fit <- lm(delta_T ~ urban_imp, data=sub); sm <- summary(fit)
    data.frame(pair=p, urban=unique(sub$urban), rural=unique(sub$rural),
               imp_range=sprintf("%.1f-%.1f", min(sub$urban_imp), max(sub$urban_imp)),
               slope=round(coef(fit)[2], 4), ci_lo=round(confint(fit)[2, 1], 4),
               ci_hi=round(confint(fit)[2, 2], 4), r2=round(sm$r.squared, 3),
               p_value=signif(coef(sm)[2, "Pr(>|t|)"], 4))
  }))
  out <- out[order(-out$slope), ]
  write.csv(out, file.path(RES, "landcover_regression.csv"), row.names=FALSE)

  # §3.9 figure -- per-pair scatter of annual ΔT_mean vs urban-side imperv %
  png(file.path(FIG, "landcover_regression.png"), 2200, 1500, res=170)
  op <- par(mfrow=c(3, 3), mar=c(3.6, 3.6, 2, 0.8), mgp=c(2.3, 0.6, 0))
  for (p in sort(unique(ann$pair))) {
    sub <- ann[ann$pair==p, c("year","urban","delta_T")]
    sub <- sub[order(sub$year), ]; sub$urban_imp <- interp(unique(sub$urban), sub$year)
    fit <- lm(delta_T ~ urban_imp, data=sub); sm <- summary(fit)
    plot(sub$urban_imp, sub$delta_T, pch=21, bg="steelblue", col="navy", cex=1.1,
         xlab=sprintf("Urban-side imperv %% (1 km buffer, %s)", unique(sub$urban)),
         ylab="Annual ΔT_mean (°C)", main=p, las=1,
         panel.first=grid(col="grey92", lty=1))
    xr <- range(sub$urban_imp)
```

```
    lines(xr, predict(fit, data.frame(urban_imp=xr)), col="firebrick", lwd=1.6)
    mtext(sprintf("slope %+0.3f °C/%%  R² %.2f  p=%.3f",
                  coef(fit)[2], sm$r.squared, coef(sm)[2, "Pr(>|t|)"]),
          side=3, line=0.1, cex=0.7, adj=0)
  }
  par(op); dev.off()
  invisible(out)
}
landcover_regression()
```

7. Suggested next steps to complete the study

The present pipeline produces a defensible 21-year UHI baseline for Vienna with quantified trends and a transparent ideal-night filter. To bring the study to publication or thesis quality, four lines of work would meaningfully strengthen the conclusions.

7.1 Investigate the Langenlebarn anomaly

The negative ideal-night ΔT trends at all three LL pairs with a 2011 Pettitt break (see Figure 6 and the bottom three blue bars of Figure 7) are the most consequential unexplained result. Three concrete actions:

- Pull the full GeoSphere metadata history for station 51 (sensor changes, siting reports, surroundings notes).
- Apply the SNHT directly to the monthly LL series with neighbour pairs that exclude LL (HW, GE, MB) to localise the inhomogeneity.
- Compare HRL Imperviousness 2006/2012/2018 in a 1 km buffer around the LL site to quantify any land-cover trend that could explain a real rural-catch-up signal (Hammerberg et al., 2018).

Decision rule: if SNHT detects a clear step at 2010-2011 in monthly LL Tmean against any rural neighbour, treat the trend as an artefact and apply a segment-wise adjustment; otherwise document it as a genuine signal and add it to the discussion.

7.2 Extend the time window backward

Twenty-one years is short for UHI-trend detection – (Böhm, 1998) needed 45 years for a +0.6 °C trend to clear the noise floor. Two extensions are realistic:

- **1991-2025 (35 years)** using the same six stations. The 1991-2020 WMO baseline period would also become natural for anomaly definitions. Requires re-running `01_download` with `start = "1991-01-01"` and re-running the homogenisation step, which will likely detect the documented TAWES-automation breakpoints around 2000.
- **1981-2025 (45 years)** using only Hohe Warte and Groß-Enzersdorf, which both have long records. This is the (Böhm, 1998) design replicated on a modern dataset and would be the most directly comparable to the published Vienna literature.

7.3 Add satellite-derived surface UHI

The station network resolves six points; the satellite record resolves Vienna at 100-1000 m. Two products are free:

- **MODIS MOD11A1/MYD11A1 LST** (daily, 1 km, 2000-present) for a continuous SUHI time series. Daytime SUHI is typically stronger than the canopy-layer UHI measured at stations.
- **Landsat 8/9 Collection 2 thermal band 10** (16-day, 100 m native) for 2-3 cloud-free composite dates per year.

Correlating the station ΔT (this report) against the satellite SUHI map at the station locations would validate that the six-station gradient captures the city-wide spatial signal.

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List of tables and figures

References

Alexandersson, H. (1986). A homogeneity test applied to precipitation data. *Journal of Climatology*, 6(6), 661-675. https://doi.org/10.1002/joc.3370060607

Bassani, F., Garbero, V., Poggi, D., Ridolfi, L., & von Hardenberg, J. (2022). The interplay between urban heat islands and heat waves in 32 European cities. *Environmental Research Letters*, 17(12), 124006. https://doi.org/10.1088/1748-9326/aca3ba

Böhm, R. (1998). Urban bias in temperature time series - A case study for the City of Vienna, Austria. *Climatic Change*, 38(1), 113-128. https://doi.org/10.1023/A:1005338514333

GeoSphere Austria. (2024). Data Hub - klima-v2-1d, klima-v2-1h, synop-v1-1h resources. https://data.hub.geosphere.at

Guijarro, J. A. (2018). *Homogenization of climatic series with Climatol* (Version 3.1.1). State Meteorological Agency (AEMET), Spain. https://CRAN.R-project.org/package=climatol

Hammerberg, K., Brousse, O., Martilli, A., & Mahdavi, A. (2018). Implications of employing detailed urban canopy parameters for mesoscale climate modelling: A comparison between WUDAPT and GIS databases over Vienna, Austria. *International Journal of Climatology*, 38, e1241-e1257. https://doi.org/10.1002/joc.5447

Hirsch, R. M., & Slack, J. R. (1984). A nonparametric trend test for seasonal data with serial dependence. *Water Resources Research*, 20(6), 727-732. https://doi.org/10.1029/WR020i006p00727

Hutter, H.-P., Poteser, M., Lemmerer, K., Wallner, P., Kundi, M., Moshammer, H., & Weitensfelder, L. (2023). Heat-related mortality in three European cities - A descriptive analysis for sustainable adaptation. *Atmosphere*, 14(10), 1498. https://doi.org/10.3390/atmos14101498

Kendall, M. G. (1975). *Rank correlation methods* (4th ed.). Charles Griffin, London.

Oke, T. R. (1982). The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455), 1-24. https://doi.org/10.1002/qj.49710845502

Pettitt, A. N. (1979). A non-parametric approach to the change-point problem. *Journal of the Royal Statistical Society Series C (Applied Statistics)*, 28(2), 126-135. https://doi.org/10.2307/2346729

Sen, P. K. (1968). Estimates of the regression coefficient based on Kendall's tau. *Journal of the American Statistical Association*, 63(324), 1379-1389. https://doi.org/10.1080/01621459.1968.10480934

Stauffer, R. (2024). *gsdata: GeoSphere Austria Data API Client* (Version 0.2-x). https://retostauffer.org/news/gsdata/

Stewart, I. D., & Oke, T. R. (2012). Local Climate Zones for urban temperature studies. *Bulletin of the American Meteorological Society*, 93(12), 1879-1900. https://doi.org/10.1175/BAMS-D-11-00019.1

Žuvela-Aloise, M., Koch, R., Buchholz, S., & Früh, B. (2016). Modelling the potential of green and blue infrastructure to reduce urban heat load in the city of Vienna. *Climatic Change*, 135(3-4), 425-438. https://doi.org/10.1007/s10584-016-1596-2